

Information Technology and Returns to Scale[†]

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What are the implications of the dramatic fall in IT prices for aggregate technology? When firm-level technologies are continuously differentiable, a factor price shock leads to (i) a substitution between factors and/or (ii) an endogenous response of returns to scale. The second channel is governed by the output elasticity of relative factor demand. Using detailed firm-level data from France, we estimate this elasticity to be positive for IT factor demand. A quantitative exercise accounting for both technological channels shows that falling IT prices can explain much of the changes in concentration and the composition of aggregate labor share in France. (JEL D22, D24, D33, E25, L63, L86)

Over the past few decades, advances in information technology (IT) have drastically lowered the quality-adjusted prices of software and computing equipment. In response, business investment in IT-intensive tools has soared globally. This rise in IT investment has fueled productivity growth both at the firm level (see, e.g., Brynjolfsson and Hitt 2003; Bloom, Sadun, and Van Reenen 2012) and at the aggregate (see, e.g., Basu et al. 2003). However, recent work suggests that a number of potential macroeconomic side effects might have accompanied these productivity gains. For instance, the fall in IT prices has been so dramatic as to drive down the overall price of equipment capital, potentially contributing to a substitution away from labor and a resulting decline in labor's share of aggregate income (Karabarbounis and Neiman 2014; Eden and Gaggl 2018).¹ The rise of IT has also

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¹A sizable body of work has examined the available evidence for a potential global fall in the labor share and the different likely explanations (e.g., Elsby, Hobijn, and Sahin 2013; Koh, Santaaulàlia-Llopis, and Zheng 2015; Barkai 2020; Grossman et al. 2017). For a recent reexamination of the contribution of IT to the fall in the price of capital, see Gourio and Rognlie (2020).

been linked to the recent surge in industrial concentration of production and the dominance of large firms across many developed economies (Autor et al. 2020).²

In this paper, we revisit the consequences of the fall in IT prices for aggregate productivity, factor distribution of income, and industry concentration in a framework that bridges micro and macro levels. Our empirical contribution is to introduce a novel and unique dataset that includes administrative records on firm-level software and hardware investments from France.³ Using these data, we document a new fact that plays a key role in shaping the economy's response to falling IT prices: factor demand is more intensive in IT inputs for French firms that operate on a larger scale. This scale dependence (nonhomotheticity) in IT demand creates a tight link between the rise of IT and that in the concentration of production among large firms. Motivated by this fact, we theoretically study the aggregation of the responses of heterogeneous firms with smooth production technologies to a change in the price of a given factor of production. Applied to the case of IT, our theory is compatible with the observed scale dependence in the data and thus contrasts with the prior aggregation theories that assume CRS and scale-invariant (homothetic) technologies (Oberfield and Raval 2021; Baqaee and Farhi 2019). We show that accounting for the scale dependence in IT demand allows us to explain a sizable part of the trends in the French economy as the response to the observed fall in IT prices.

We begin by providing evidence for scale dependence among French firms, showing that larger firms use IT more intensively compared to smaller firms and hence, disproportionately benefit from falling IT prices. The positive relationship between scale and the IT intensity of inputs is robust to measuring IT as flows or stocks of investment for either software or hardware, and to measuring intensity relative to labor inputs, non-IT capital, or total cost. The relationship is also robust to different measures of firm scale, such as employment, value added, and sales.⁴ We find the relationship to hold within and across firms and to remain stable as we move from firms hiring just a few workers to those hiring thousands. It also holds if we estimate the (within-firm) output elasticity of relative IT demand using firm-level export demand shocks as instruments for output in the sample of exporting firms.

We next explore the theoretical consequences of such scale dependence in IT demand. We show that the output elasticity of relative demand for a given factor plays a key role in shaping the response of price, output, and income shares to the fall in the price of that factor. When the demand for a factor is more elastic in output compared to other factors, a fall in its price raises the returns to scale at the firm level.

²A number of papers have documented a rise in the industrial concentration of production and within-industry dispersion in firm outcomes in the United States, across Europe, and other OECD countries (e.g., Andrews, Criscuolo, and Gal 2016; Autor et al. 2020; Berlingieri, Blanchenay, and Criscuolo 2017; CEA 2016; Bajgar et al. 2019; Kwon, Ma, and Zimmermann 2022).

³Empirical studies of the macroeconomic consequences of the rise of IT have been hampered by the lack of availability of comprehensive data on firm-level IT investments. Available data have been limited to business surveys, such as the CII or Harte-Hanks (e.g., Bresnahan, Brynjolfsson, and Hitt 2002; Bloom, Sadun, and Van Reenen 2012), or limited surveys conducted by government agencies, such as the European community surveys (e.g., Schivardi and Schmitz 2020). More recently, the US Census Bureau has begun collecting comprehensive, nationally representative data on firm usage of advanced technologies including IT as part of the Annual Business Survey (ABS) starting in 2018 (see Zolas et al. 2020 for details). We note that the software module of our data has been subsequently used by De Ridder (2024).

⁴We also show that the relationship between size and IT intensity holds even if we rely on more eclectic proxies for firm size, such as the number of plants, the depth of organizational structure, the number of exporting markets, or the number of exported products (the latter two measures only in the sample of exporting firms).

This *endogenous response in returns to scale* leads to a channel that is conceptually distinct from the standard substitution channel that lies at the core of the standard theories of aggregate technology. Just like this standard channel, the returns to scale channel has important consequences for the factor shares of income. Even if we consider constant markups, the share of profits in firm income endogenously varies with returns to scale.⁵ In particular, if the output elasticity of relative IT demand is positive, a fall in IT prices reduces profit shares as it raises returns to scale. Moreover, under certain conditions (including gross complementarity between IT and other factors), firms with high IT intensity in the cross section are predicted to have lower returns to scale and higher profit shares. Such variations in profit shares over time and across firms in turn lead to variations in labor share that are independent of the degree of substitutability between labor and IT.

We also provide a model that extends these insights from the firm to the aggregate level, considering a collection of firms with heterogeneous, continuously differentiable cost functions. To focus attention on the role of technology, we assume competitive factor markets and a standard product market structure featuring CES product demand and monopolistic competition. We derive analytical expressions for (i) the aggregate elasticity of substitution between IT and other factors and (ii) the aggregate output elasticity of aggregate relative IT demand, as a function of the firm-level technological elasticities and the distribution of production across firms. We find that, due to between-firm reallocation effects, macro- and micro-level patterns of scale dependence may diverge, potentially leading to scale invariance in aggregate technology with scale-dependent firm-level technology, and vice versa.⁶

Using this model, and under simple assumptions about factor supplies, we further derive the general equilibrium response of the economy to falling IT prices, accounting both for within-firm and for between-firm reallocation responses. The presence of scale dependence in IT demand impacts both these responses. For instance, a fall in IT prices shifts the production toward firms with high IT intensities. Since, as mentioned, these firms may operate with higher profit shares, this reallocation effect can raise the aggregate profit share and lower the aggregate labor share, counteracting the within-firm response. Moreover, the magnitude of the predicted reallocation response is also influenced by the endogenous rise in returns to scale at the firm level. For instance, the returns to scale channel implies that falling IT prices lowers marginal costs more than average costs. This effect tends to increase the pass-through of the IT prices into the marginal cost (and output prices). Finally, when aggregate technology is scale dependent, the response of aggregate output may lead to additional responses in prices, factor shares, and the concentration of production. All

⁵The ratio of profits to costs is given by $(P \cdot Y - C)/C = \mu(Y\partial C/\partial Y) - 1$, which is a linear function of the cost elasticity $Y\partial C/\partial Y$ assuming constant markups μ . Here, variations in the share of profits in income do not stem from differences in market power. Instead, they correspond to differences in Ricardian rents, i.e., the returns to fixed, firm-specific, inimitable factors that distinguish firms from one another. Since we assume that all other factors receive their marginal products, the residual value accrues to the owner of the firm.

⁶In fact, we rely on the latter case to provide a microfoundation for scale dependence for a firm that has access to alternative scale-invariant production techniques with varying degrees of returns to scale. If the returns to scale and the IT intensities positively covary across these techniques, a rise in firm output shifts activity toward techniques with higher returns to scale and thus also raises IT intensity (hence, scale dependence). Likewise, a fall in IT price shifts activity toward techniques with higher IT intensity and thus also raises returns to scale (hence, the returns to scale channel).

in all, our results allow us to fully characterize the general equilibrium response of the macroeconomy to a change in factor prices, based on micro-level technological elasticities as well as the distribution of production across firms. While the latter can be observed in our micro data, we need to estimate the former at the firm level.

To estimate these elasticities, we consider a nested *Nonhomothetic CES* (nhCES) firm-level production function that combines IT and non-IT inputs (Sato 1974, 1977; Comin, Lashkari, and Mestieri 2021). This specification assumes that our two main technological elasticities, i.e., (i) the elasticity of substitution between IT demand and other factors and (ii) the output elasticity of IT demand relative to other factors, are two distinct and constant structural parameters. These two elasticities in turn characterize the two main channels, substitution and returns to scale, respectively. We adapt the standard production function estimation framework to a setting with factor-augmenting productivity shocks using an approach similar to Doraszelski and Jaumandreu (2018). In addition to the common assumptions on the timing of firm decisions, we further rely on shift-share instruments for the price of software relative to wages to help estimate the elasticity of substitution (similar to Oberfield and Raval 2021). Confirming the reduced-form evidence, we find a sizable output elasticity of relative IT demand and an elasticity of substitution between IT and non-IT inputs suggesting gross complementarity (below unity).⁷ Finally, our estimated parameters imply values for the degree of returns to scale that are close to unity for the average firm, while still leading to sizable heterogeneity in the cross section depending on the firm's IT intensity.

Armed with these estimates, we rely on our aggregation results to study the predictions for the aggregate consequences of the fall in IT prices. Official statistics suggest a substantial fall in IT prices and a concurrent rise in the IT intensity of aggregate capital in France over the 1995–2007 period. Since the official statistics are likely to underestimate the degree of price declines due to inadequate quality adjustments, we also consider alternative scenarios with faster price declines. Using our firm-level data, we further document a rise in the industrial concentration of production, as well as a rise in labor share within firms that is canceled in the aggregate by reallocations toward low-labor share firms.⁸ Our framework predicts a rise in industry concentration that is qualitatively in line with and quantitatively close to the magnitude of the patterns observed in the data.

Prior Work.—Our paper contributes to a large literature that has studied the impacts of IT at both the micro and the macro levels.⁹ We also contribute to an

⁷We find similar estimates in the sample of all industries, the sample of manufacturing industries, and in samples of more disaggregated industries. The estimated output elasticity of relative IT demand and elasticity of substitution are 0.43 and 0.28 (0.47 and 0.13) in the sample of all (manufacturing) industries. These numbers are close to, but somewhat smaller than, recent micro-level estimates of the elasticity of substitution between capital and labor (see, e.g., Oberfield and Raval 2021; Doraszelski and Jaumandreu 2018).

⁸The simultaneous rise in within-firm and fall in across-firm contributions to the aggregate shares has already been documented in the case of the United States (e.g., Autor et al. 2020; Kehrig and Vincent 2021). While the two effects cancel out in France, the net effect appears to have been negative at least in the case of US manufacturing.

⁹Other contributions to the extensive empirical literature on the productivity effects of IT include, for micro-level studies, Brynjolfsson and Yang (1996); Brynjolfsson and Hitt (2003); Aral, Brynjolfsson, and Wu (2006); Draca, Sadun, and Van Reenen (2009); Bloom, Sadun, and Van Reenen (2012), and for macro-level ones, Oliner and Sichel (2000); Jorgenson (2001); Stiroh (2002, 2008); and Jorgenson, Ho, and Stiroh (2005). For a number of recent empirical contributions, see, e.g., Caroli and Van Reenen (2001); Akerman, Gaarder, and Mogstad

already extensive literature that studies the recent rise in industry concentration, and a number of other secular macroeconomic trends across advanced economies.¹⁰ More specifically, a number of recent contributions, including Aghion et al. (2023); Mariscal (2018); and De Ridder (2024) have also linked the rise of IT, or intangible capital more broadly, to the recent secular macroeconomic trends in the United States and across other countries. We complement these theoretical studies by grounding our analysis in detailed micro-level data and performing a quantitative aggregation of the micro-level responses to a well-defined shock, implied by the observed fall in IT prices. Moreover, our approach is distinct from the aforementioned papers in another key dimension. While they each consider the effects of IT through the combination of a specific technological mechanism along with market power, our approach provides a comprehensive account of all such *purely technological forces* at the level of the firm production function.¹¹ We will provide a more detailed discussion of these points in Section VB.

The remainder of this paper is organized as follows. Section I presents the data sources and the reduced-form evidence on the relationship between firm size and relative IT demand. Section II presents the theory. Section III discusses the structural estimation of firm-level technology. Section IV discusses the quantitative results of the calibration. Finally, Section V provides additional discussions of our framework and results, and Section VI concludes the paper.

I. Motivating Fact: Firm Scale and Relative IT Demand

In this section, we introduce our data sources and use them to document a positive relationship between firm scale and the relative IT demand, across and within firms.

A. Data

Our data cover active firms in the corporate sector in France from 1989 to 2007, collected by the French Institute of Statistics and Economic Studies (INSEE) from firm-level surveys and tax records as well as employee-level records and product-level customs records. It provides standard income and employment information for all French firms that report to the tax and social security authorities, as well as their investments in several distinct classes of assets.

The main novelty of our data is to include detailed information on investments in software and hardware of firms above certain thresholds of size. The annual survey of firms (Enquête annuelle d'entreprise, EAE) files (INSEE 2007a) contain

(2015); DeStefano, Kneller, and Timmis (2014, 2018); Harrigan, Reshef, and Toubal (2016, 2018); Gaggl and Wright (2017); and Acemoglu et al. (2014).

¹⁰ Alternatively, Crouzet and Eberly (2017, 2018); Gouin-Bonenfant (2022); Martinez (2018); Akcigit and Ates (2023); Hopenhayn, Neira, and Singhanian (2018); and Liu, Mian, and Sufi (2019) have proposed the rise of intangible capital, productivity dispersion, automation, and the decline in knowledge diffusion, population growth, and interest rates as potential drivers for the recent macroeconomic trends, respectively.

¹¹ While in this paper we do not study the within-firm drivers of these forces, the theories of the organization of production within firm may help rationalize patterns similar to those uncovered here (Caliendo and Rossi-Hansberg 2012; Caliendo, Monte, and Rossi-Hansberg 2015; Caliendo et al. 2020). For a direct application of this theory to models of IT and firm organization, see Bloom et al. (2014) or Mariscal (2018). Relatedly, Hsieh and Rossi-Hansberg (2023) provide a theory that links IT to the trends in local labor market concentration.

information on software investment for all firms with more than 20 employees and a representative sample of smaller firms (Frouté 2001), and the normal tax regime (Bénéfices réels normaux, BRN) files (INSEE 2007b) report the total investment on office and computing equipment of firms with sales above a certain threshold (Barbesol, Heckel, and Quantin 2008; Chevalier and Luciani 2018).¹²

We construct firm-level measures of the stocks of hardware, software, and other types of non-IT capital, applying the perpetual inventory method (PIM) to the observed values of investment, deflated by the corresponding price series. We start this procedure from an initial stock in 1989 but discard the first five years of the stock data, focusing on the period 1995–2007, to ensure that our results are not driven by the initial values of stocks. We also build distinct measures for the rental prices of each asset type in the data (software, hardware, and the different subclasses of non-IT capital).¹³

B. Reduced-Form Evidence

We first investigate the within-industry relationship between IT per employee and the scale of production across firms by estimating the equation

$$(1) \quad ITperemployee_{it} = FE_{Size_{it}} + FE_{ind_{it}} + FE_{age_{it}} + FE_{coh_{it}} + \nu_{it},$$

where $ITperemployee_{it}$ denotes a measure of IT demand (investments or stocks of software or hardware) relative to the number of employees, $FE_{Size_{it}}$ denotes a size-class dummy corresponding to the firm i 's number of employees at time t , and $FE_{ind_{it}}$ stands for a flexible set of industry-time fixed effects (at the three-digit level). In addition, we further add age $FE_{age_{it}}$ and cohort $FE_{coh_{it}}$ dummies to flexibly control for the effect of age and cohort.¹⁴

Figure 1 shows the estimated coefficients for different size classes. In all cases, the estimates are reported relative to the estimated dummy for the size class corresponding to zero to five employees. We find a strong relationship between firm size and software and hardware per employee, both in terms of investment and stock. For instance, a typical firm with more than 5,000 employees invests close to €200 (€250) per employee more in software (hardware) than firms with 0–5 employees.

The IT *premium* of larger firms is fairly sizable considering that the unweighted mean software (hardware) per employee across all firms is around €28

¹²In 2007, the thresholds were €763,000 if the firm operates in trade or real estate sectors, and €230,000 otherwise. Files (INSEE 2007c) with information on small firms that report to the simplified regime (Régime simplifié d'imposition, RSI) instead of the normal BRN tax regime and are never surveyed by EAE are included in the analysis of the macro trends in France in Section IV, but are not included in the IT data. See more details about our sources of micro data in Section OA.2.4 of the online Appendix.

¹³For this procedure, we employ asset-specific price deflators, depreciation rates, and initial industry-level stocks in year 1990 from aggregate national accounts data (INSEE 2019; Eurostat 2019) and KLEMS data (Conference Board 2023a,b; Stehrer et al. 2019). We employ financial data to compute the cost of capital (OECD 2020b,a; Banque de France 2019, 2020; INSEE 2020). See more details about our depreciation rates, price deflators, industry-level stocks, our cleaning procedure, and our perpetual inventory procedure in Sections OA.2.2.2, OA.2.2.3, OA.2.3, OA.2.5, and OA.2.6 of the online Appendix, respectively. Table OA.1 in the online Appendix presents the summary statistics of the main variables in our data.

¹⁴Since it is not possible to separately identify the age, cohort, and year fixed effects, we apply the normalization suggested by Deaton (2018), but the results are also similar without these controls. See the details of this normalization and results without these controls in Section OA.3.1.3 and Table OA.28 of the online Appendix.

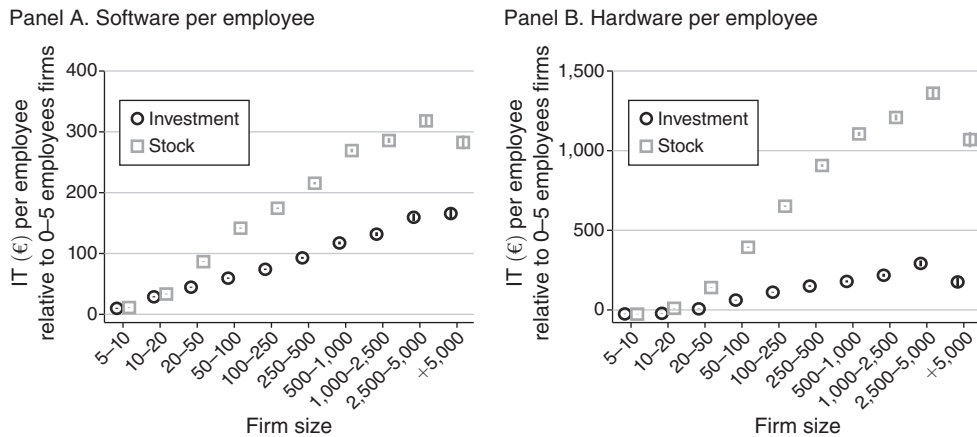


FIGURE 1. CROSS-SECTIONAL RELATIONSHIP BETWEEN IT PER EMPLOYEE AND FIRM SIZE

Notes: This figure reports average IT investment and stock per employee by firm size class. Averages are conditional on a set of flexible fixed effects constructed from the interaction of three-digit industry codes and time dummies, and a full set of cohort fixed effects (pre-1980, 1980–1993, 1993–1995, . . . , 2005–2007) and normalized age fixed effects. In the case of software, the sample is all firms that were sampled in EAE (that year for investment, at least once for capital). In the case of hardware, the sample is all firms that reported hardware investment lower than 0.99 times total investment. The bands around the estimates show the 90 percent confidence intervals.

(€178).¹⁵ As the figure shows, the results are robust with respect to using stocks rather than investments. Moreover, as we show in Appendix B, the results are also robust with respect to using measures of IT demand relative to other factors, such as IT per unit of capital or IT per unit of total costs.

Next, we measure firm scale in terms of output (instead of employment) and estimate the IT demand equation

$$(2) \quad \log(ITperemployee_{it}) = \eta \log Size_{it} + FE_{ind_{it}} + FE_{age_{it}} + FE_{coh_{it}} + \nu_{it},$$

where $Size_{it}$ denotes firm value added or revenue. The first two columns of Table 1 show the coefficients η corresponding to the measures IT demand relative to the number of employees in terms of the stocks of software and hardware. The coefficients are in the 0.4–0.6 range, suggesting that raising the scale of firm output by a factor of 2 raises its stock of IT per employee by about 30–50 percent, with elasticities similar for software and hardware. As we show in Appendix B, the relationship remains fairly stable across different brackets of firm size.¹⁶

We can also examine the within-firm relationship between firm scale and IT by including a firm fixed effect FE_i in the estimation equation (2). Columns 3 and 4 of

¹⁵Note that the results in Figure 1 include the variations in relative IT demand both along the intensive and the extensive margins. The results in Table 1 instead focus only on the intensive margin of relative IT demand. Figure OA.10 in the online Appendix focuses only on the extensive margin of investment and shows that the likelihood of investing in IT grows in firm size for both software and hardware.

¹⁶We provide additional details and results in Section OA.3.1 of the online Appendix. Online Appendix Table OA.3 presents the full set of results using alternative proxies for IT intensity. Online Appendix Table OA.4 shows a similar relationship between IT intensity and a number of other proxies of firm scale, including the number of plants, the depth of the organizational structure, the number of exporting products, and destinations. Section OA.3.2 of the online Appendix provides similar details for the within-firm estimates.

TABLE 1—REGRESSIONS OF LOG IT PER EMPLOYEE ON LOG FIRM SIZE

	Within-industry		Within-firm	
<i>Panel A. Software (stock) per employee</i>				
Size (proxied by sales)	0.4131 (0.0033)		0.2138 (0.0328)	
Size (proxied by VA)		0.4092 (0.0036)		0.2224 (0.0289)
Observations	556,996	557,059	236,510	236,617
<i>Panel B. Hardware (stock) per employee</i>				
Size (proxied by sales)	0.6249 (0.0013)		0.2681 (0.0097)	
Size (proxied by VA)		0.5780 (0.0014)		0.1520 (0.0082)
Observations	1,299,964	1,300,197	249,933	250,935

Notes: In both panels the dependent variable is the logarithm of IT stock per employee. Standard errors are reported in parentheses. The independent variable is the logarithm of firm size either proxied by sales or value added. The time period is 1995–2007. The sample is firms with at least ten employees, in panel A, sampled by EAE, and in panel B, BRN firms. All columns include a full set of three-digit industry classification fixed effects interacted with year fixed effects, and within-industry results also include a full set of cohorts fixed effect (pre-1980, 1980–1993, 1993–1995, . . . , 2005–2007) and normalized age fixed effects. An elasticity of 0.4 means that raising sales by a factor of 2 raises IT per employee by 40 percent. Within-firm standard errors are clustered at the level of the firm.

Table 1 present the coefficients of firm size in the same regressions as before that now additionally include firm fixed effects. We find positive and significant coefficients similar to the cross-sectional estimates, in the 0.15–0.25 range.

The facts presented so far suggest the potential presence of scale dependence in IT demand. After exploring the theoretical implications of such scale dependence in Section II, we will present a structural approach to the identification of the size elasticity of relative IT demand in Section IIIB. Additionally, Section OA.3.3 of the online Appendix presents an alternative, reduced-form approach to the identification of this elasticity. Therein, we discuss how the two structural and reduced-form approaches are consistent with one another, reinforcing the evidence in favor of scale dependence in firm-level demand for IT.

II. Theory: Technological Response to Factor Price Shocks

In this section, we examine the theoretical implications of scale-dependent (non-homothetic) factor demand. Even though our focus is ultimately on IT factor demand, the theory in this section more generally studies the shock to the price of any factor of production exhibiting scale-dependent demand. We begin in Section IIA by revisiting the production theory and show that scale-dependent factor demand implies a novel channel through which a factor price shock propagates to firm-level prices and affects factor shares of income. We then apply this insight in Section IIB to a model that aggregates firm-level responses to factor price shocks to study the implications for the responses of industry concentration, factor distribution of income, and output in the aggregate.

A. Scale-Dependent Factor Demand and Returns to Scale

Consider a production function $Y = \mathcal{F}(\mathbf{X})$ that transforms a vector of J factors $\mathbf{X} \in \mathbb{R}_+^J$ into output Y . Cost minimization allows us to define a corresponding firm-level cost function $C \equiv \mathcal{C}(Y; \mathbf{W})$ for given vector of factor prices $\mathbf{W} \in \mathbb{R}^J$, which we assume to be third-order continuously differentiable. Below, we begin by studying the main elasticities of the production technology that characterize its dependence on factor price shocks or on the scale of output.

The Elasticities of Cost and Factor Demand.—Let us define two elasticities of the cost function with respect to the price of factor j and with respect to output as

$$(3) \quad \Omega_j(Y; \mathbf{W}) \equiv \frac{\partial \log \mathcal{C}(Y; \mathbf{W})}{\partial \log W_j},$$

$$(4) \quad \mathcal{E}(Y; \mathbf{W}) \equiv \frac{\partial \log \mathcal{C}(Y; \mathbf{W})}{\partial \log Y}.$$

Equation (3) characterizes factor demand since, by Shephard's lemma, it equals the share of factor j in costs, $\Omega_j = W_j X_j / C$. The elasticity in equation (4) characterizes the degree of returns to scale since it is the reciprocal of the scale elasticity, exceeding unity under decreasing returns and falling short of unity under increasing returns.¹⁷ Henceforth, to simplify the exposition, we refer to the elasticities Ω_j as the *factor intensity* (of costs) and to the elasticity $\mathcal{E}$ as the *cost elasticity*.¹⁸

By definition, cost elasticity is the ratio of marginal cost $\mathcal{C}'(Y; \mathbf{W}) \equiv (\partial \mathcal{C} / \partial Y)(Y; \mathbf{W})$ to average cost C/Y . Thus, the product of cost elasticity $\mathcal{E}$ and markup μ gives us the gap between price and average cost, between the income (revenue) and total cost, and between the factor share (share of the factor in income) and factor intensity:

$$(5) \quad P = \mu \mathcal{E} \frac{C}{Y}, \quad \frac{PY}{C} = \mu \mathcal{E}, \quad FS_j \equiv \frac{W_j X_j}{PY} = \frac{\Omega_j}{\mu \mathcal{E}}.$$

As equation (5) shows, cost elasticity has a similar role to that of markup in shaping price and factor shares. Variations in markup and in returns to scale (cost elasticity) are two alternative mechanisms for changing the share of profits in income. The former indicates differences in market power, whereas the latter stems from variations in Ricardian rents and does not involve distortive allocations.

We next define two elasticities of factor intensity (factor demand) with respect to factor price and with respect to output. The elasticity of substitution between factor

¹⁷Formally, the scale elasticity is defined as $\lim_{b \rightarrow 1} \partial \log \mathcal{F}(b\mathbf{X}) / \partial \log b$ and is equal to $1/\mathcal{E}(Y; \mathbf{W})$ for the vector of factor demand $\mathbf{X}$ corresponding to output Y and the price vector $\mathbf{W}$.

¹⁸The term "cost elasticity" (or the elasticity of cost) has been previously used in the literature on production theory (see, e.g., Hanoch 1975a; Brown and Chachere 1980, 1986). Frisch (1964) instead uses the term "substitutable cost flexibility."

j and all the other factors (σ_j) and the output elasticity of the relative demand for this factor (η_j) are given by¹⁹

$$(6) \quad 1 - \sigma_j(Y; \mathbf{W}) \equiv \frac{1}{1 - \Omega_j(Y; \mathbf{W})} \frac{\partial \log \Omega_j(Y; \mathbf{W})}{\partial \log W_j},$$

$$(7) \quad \eta_j(Y; \mathbf{W}) \equiv \frac{1}{1 - \Omega_j(Y; \mathbf{W})} \frac{\partial \log \Omega_j(Y; \mathbf{W})}{\partial \log Y}.$$

The elasticity of substitution measures how much factor intensity changes when the price of a factor changes. Similarly, the output elasticity of relative factor demand measures how much factor intensity changes when the scale of output changes. Under standard scale-invariant (homothetic) technologies, e.g., all CRS technologies, the composition of factor inputs does not depend on the scale of production, and this elasticity is zero, $\eta_j \equiv 0$. Under more general technologies, however, the scale of production may impact the composition of factor demand. In particular, our results in Section IB suggest that relative IT demand is best characterized by a positive output elasticity $\eta_I > 0$.

The Effects of Factor Price Shocks: The Returns to Scale Channel.—We now show that scale-dependent factor demand implies a new channel through which changes in the factor price propagate to the price of output and to the factor's share of income. Equation (5) shows that the effect of any shock on output prices and on factor shares is mediated by its impact on the cost elasticity (and markups), as it is by its impacts on average costs and factor intensities. To focus on the technological mechanisms, let us assume markups are constant and exogenous and focus on the response of the cost elasticity to a factor price change. The next lemma shows that the elasticities of the cost function and factor demand, defined above, together characterize this response.

LEMMA 1: *The elasticity of cost elasticity with respect to the price of IT is related to the output elasticity of relative IT demand through*

$$(8) \quad \frac{\partial \log \mathcal{E}}{\partial \log W_j} = \frac{1 - \Omega_j}{\mathcal{E}} \Omega_j \eta_j.$$

PROOF:

Using the definition of cost elasticity (4), we have $\partial \mathcal{E}_j / \partial \log W_j = \partial^2 \log \mathcal{C} / (\partial \log W_j \partial \log Y) = \partial \Omega_j / \partial \log Y$, where in the second equality we have used the definition of factor intensity (3). ■

¹⁹The output elasticity of relative factor demand is related to the output elasticity of factor demand through $\partial \log X_j(Y; \mathbf{W}) / \partial \log Y = (1 - \Omega_j) \eta_j + \mathcal{E}$. We choose to specify the output elasticity of relative factor demand since it is in line with our empirical specification in Section I. Moreover, as we will see in the specification of nhCES production functions in Section IIIA, the elasticity of relative factor demand can be a constant primitive under scale-dependent (nonhomothetic) technology.

Although Lemma 1 may seem simple, it offers an important insight: the degree of scale dependence in the relative demand for a given factor determines how changes in the price of that factor affect returns to scale. For instance, in Section IB, we showed that IT intensity tends to increase with firm size, suggesting that $\eta_I > 0$ when considering IT as a factor ($j = I$). Accordingly, the lemma implies that a fall in IT prices tends to lower marginal costs more than average costs, decreases the cost elasticity, and increases returns to scale.

This endogeneity in returns to scale matters for the responses of price and factor shares. Since markups are constant, Lemma 1 and equation (5) together imply that the pass-through of a change in the price of factor j into the marginal cost (and price) and the effect of this factor price shock on the income share of other factors are given by

$$(9) \quad \frac{\partial \log P}{\partial \log W_j} = \Omega_j^m = \Omega_j \left(1 + \frac{1 - \Omega_j}{\mathcal{E}} \eta_j \right),$$

$$(10) \quad \frac{\partial \log FS_{-j}}{\partial \log W_j} = \Omega_j \left(\sigma_j - \frac{1 - \Omega_j}{\mathcal{E}} \eta_j - 1 \right),$$

where we have used FS_{-j} to indicate the share of all other factors in income. Conventional specifications of technology commonly assume scale invariance (homotheticity, $\eta_j = 0$). This implies that the effect of factor price shocks on output price only depends on factor intensity Ω_j , and the effect on the income share of other inputs only depends on factor intensity Ω_j and the elasticity of substitution σ_j . The elasticity of substitution measures the extent of the *substitution channel*, i.e., the endogenous response of the relative mix of factor inputs to factor price shocks.

Under more general production technologies, equations (9) and (10) show that two additional production elasticities matter for the effects of a factor price shock on the output price and on the factor shares of income: the cost elasticity $\mathcal{E}$ and the output elasticity of relative factor demand η_j . When the factor price W_j changes, the elasticity of substitution σ_j reveals how the factor intensity Ω_j responds. Similarly, the output elasticity of relative factor demand η_j reveals how the cost elasticity $\mathcal{E}$, and thus returns to scale, responds to this change. For this reason, we refer to the latter mechanism as the *returns to scale channel*.

In terms of magnitude, compared to the benchmark of scale-invariant production, the price pass-through rises by a factor of $[(1 - \Omega_j)/\mathcal{E}] \eta_j$, and the effect on the share of other factors in income is reduced by $[(1 - \Omega_j)/\mathcal{E}] \eta_j$. In the case of IT, we will empirically show that the factor intensity of IT is typically fairly small and the production is close to CRS, implying that to the first-order of approximation in Ω_I , equations (9) and (10) lead approximately to $\Omega_I(1 + \eta_I)$ and $\Omega_I(\sigma_I - \eta_I - 1)$. Thus, the pass-through is greater and the (effective) substitutability is smaller by roughly the value of the output elasticity η_I , which we will also estimate to be sizable and positive.

So far, we have examined how technology responds to a shock in factor prices at the level of a single production unit. However, if that unit is a firm and we want to understand how the shock impacts the aggregates, we need to consider how the allocation of production changes in response to such shocks, i.e., how inputs move

toward or away from the firm under consideration. The response of allocation is determined by the response of prices and hence marginal costs. For this reason, it is important to distinguish the elasticities of cost, defined in equations (3) and (4), from the elasticities of marginal cost $C' = \mathcal{E}C/Y$ with respect to factor price and output

$$(11) \quad \Omega_j^m \equiv \frac{\partial \log C'(Y; \mathbf{W})}{\partial \log W_j}, \quad \mathcal{E}^m \equiv \frac{\partial \log C'(Y; \mathbf{W})}{\partial \log Y}.$$

We have already encountered the expression for the factor price elasticity of marginal cost Ω_j^m in equation (9). The output elasticity of marginal cost $\mathcal{E}^m$ satisfies $\mathcal{E}^m = \mathcal{E} - 1 + \mathcal{E}^s$, where we have defined the elasticity of cost elasticity with respect to output, the cost superelasticity, as $\mathcal{E}^s \equiv \partial \log \mathcal{E} / \partial \log Y$.²⁰ Under the common assumption that the returns to scale is exogenous and constant, the elasticities of cost and marginal cost become tightly linked: $\Omega_j^m = \Omega_j$ and $\mathcal{E}^m = \mathcal{E} - 1$. In Section IIB, we will rely on the elasticities of marginal cost defined in equation (11) to characterize the reallocations of production across heterogeneous firms in response to factor price shocks.

B. Aggregation Model

Consider now an economy with a continuum mass N of heterogeneous, monopolistically competitive, single-product firms. We assume firm-specific production functions $Y_i = \mathcal{F}_i(\mathbf{X}_i)$ and corresponding firm-level cost functions $C_i \equiv \mathcal{C}_i(Y_i; \mathbf{W})$. We assume that the consumer demand for firm-level products is characterized by a CES aggregator with elasticity of substitution λ , implying constant markups $\mu \equiv \lambda/(\lambda - 1)$. The following assumption states our baseline assumptions about factor inputs and markets, which we will use for the derivation of the theoretical results presented in this section.

ASSUMPTION 1 (Benchmark Setting for Factor Input Decisions/Markets): *All factor inputs are “flexible,” in the sense that firms do not face adjustment costs or other frictions in their input choices. All factor markets are perfectly competitive, and firms face identical, national factor prices.*

Applying the definitions in Section IIA to the firm-level cost function $\mathcal{C}_i(Y; \mathbf{W})$, we can characterize the technology for each firm i through, firstly, the elasticities of firm-level costs $\Omega_{j,i}$ and $\mathcal{E}_i$ from equations (3) and (4); secondly, the elasticities of factor demand $\sigma_{j,i}$ and $\eta_{j,i}$ from equations (6) and (7); and thirdly, the elasticities of marginal cost $\Omega_{j,i}^m$ and $\mathcal{E}_i^m$ from equation (11). From equation (5), firm-level price, income to cost ratio, and factor shares satisfy $P_i = \mu \mathcal{E}_i C_i / Y_i$, $P_i Y_i / C_i = \mu \mathcal{E}_i$, and $FS_j = \Omega_{j,i} / \mu \mathcal{E}_i$, respectively.

²⁰In addition to the factor intensity Ω_j , cost elasticity $\mathcal{E}$, and the elasticities of factor demand σ_j and η_j , the cost superelasticity is the last elasticity that is needed to characterize the technological responses to a factor price shock. For further discussion of this elasticity, see Appendix A.A.

Let us now characterize the aggregate technology under this model. Given the assumption that markups are constant across firms, the market allocation of factors across firms is efficient. Thus, we define an aggregate cost function in terms of the aggregate output Y and factor prices $\mathbf{W}$:

$$(12) \quad \bar{C}(\bar{Y}; \mathbf{W}) \equiv \min_{(Y_i)} \int C_i(Y_i; \mathbf{W}) di, \quad \text{such that} \quad \bar{Y} = \left(\int Y_i^{\frac{\lambda-1}{\lambda}} di \right)^{\frac{\lambda}{\lambda-1}}.$$

Let the function $Y_i = \mathcal{Y}_i(\bar{Y}; \mathbf{W})$ denote the allocation of output to firm i that achieves the minimum cost for aggregate output $\bar{Y}$ and factor prices $\mathbf{W}$. This allocation also coincides with the output of firm i under a market equilibrium with the same levels of aggregate output and factor prices.²¹ The distribution of revenues across firms under this equilibrium is given by

$$(13) \quad \Lambda_i^r \equiv \frac{P_i Y_i}{\bar{P} \bar{Y}} = \left(\frac{Y_i}{\bar{Y}} \right)^{\frac{\lambda-1}{\lambda}},$$

where $\bar{P} \equiv \left(\int P_i^{1-\lambda} di \right)^{1/(1-\lambda)}$ stands for the corresponding CES aggregate price index.

Applying again the definitions in Section IIA, this time to the aggregate-level cost function $\bar{C}(\bar{Y}; \mathbf{W})$, we can characterize the aggregate technology through, firstly, the elasticities of aggregate cost $\bar{\Omega}_j$ and $\bar{\mathcal{E}}$ from equations (3) and (4); secondly, the elasticities of factor demand $\bar{\sigma}_j$ and $\bar{\eta}_j$ from equations (6) and (7); and thirdly, the elasticities of marginal cost $\bar{\Omega}_j^m$ and $\bar{\mathcal{E}}^m$ from equation (11). From equation (5), aggregate-level price index, income to cost ratio, and factor shares satisfy $\bar{P} = \mu \bar{\mathcal{E}} \bar{C} / \bar{Y}$, $\bar{P} \bar{Y} / \bar{C} = \mu \bar{\mathcal{E}}$, and $\bar{F} \bar{S}_j = \bar{\Omega}_j / \mu \bar{\mathcal{E}}$, respectively.

Aggregating Technology: From Micro to Macro.—Our first set of aggregation results characterizes aggregate elasticities of technology in terms of the firm-level elasticities. First, we can show that the two elasticities of the aggregate cost function (aggregate factor intensity and cost elasticity) equal the weighted means of the corresponding firm-level elasticities:²²

$$(14) \quad \bar{\Omega}_j = E^c[\Omega_{j,i}] \equiv \int \Lambda_i^c \Omega_{j,i} di, \quad \bar{\mathcal{E}} = E^c[\mathcal{E}_i] \equiv \int \Lambda_i^c \mathcal{E}_i di,$$

where the means are computed under the distribution implied by the shares of firm-level costs in the aggregate, given by

$$(15) \quad \Lambda_i^c \equiv \frac{C_i}{\bar{C}} = \frac{\bar{\mathcal{E}}}{\mathcal{E}_i} \Lambda_i^r.$$

The relationship between the cost and revenue distributions in equation (15) is implied by $C_i = P_i Y_i / \mu \mathcal{E}_i$ and $\bar{C} = \bar{P} \bar{Y} / \mu \bar{\mathcal{E}}$.

²¹ This allocation is characterized by the first-order condition $(1 - 1/\lambda) \bar{P} \bar{Y}^{1/\lambda} Y_i^{-1/\lambda} = \mathcal{E}_i C_i / Y_i$ and the second-order condition $\mathcal{E}_i^m > -1/\lambda$ (see Section OA.1.2 of the online Appendix).

²² See the proof of equations (14) and (15) in Section OA.1.2 of the online Appendix.

Similarly, we can express the two elasticities of the aggregate marginal cost function as weighted means of the corresponding elasticities of firm-level marginal cost

$$(16) \quad \bar{\Omega}_j^m = E^p[\Omega_{j,i}^m] = \int \Lambda_i^p \Omega_{j,i}^m di, \quad \bar{\mathcal{E}}^m = E^p[\mathcal{E}_i^m] = \int \Lambda_i^p \mathcal{E}_i^m di,$$

where the means are computed under a distribution implied by the pass-through of aggregate to firm-level output shocks, given by

$$(17) \quad \Lambda_i^p \equiv \frac{\partial \log \mathcal{Y}_i(\bar{Y}; \mathbf{W})}{\partial \log \bar{Y}} \Lambda_i^r = \frac{1 + \lambda \bar{\mathcal{E}}^m}{1 + \lambda \mathcal{E}_i^m} \Lambda_i^r.$$

The pass-through distribution reveals how an increase in the scale of aggregate output is allocated across firms, depending on their respective degrees of marginal cost elasticities. If all firms have identical marginal cost elasticities $\mathcal{E}_i = \mathcal{E}$, or if firm outputs are not substitutable $\lambda = 0$, then this allocation is proportional to the firm revenues. Otherwise, firms with higher marginal cost elasticities will receive a relatively smaller share of the overall rise in the aggregate scale.²³ Equation (17) shows that the pass-through distribution is the relevant one for aggregate marginal cost elasticities across firms.

Using the above definitions, and defining an additional distribution Λ_i^v across firms with firm-level weights given by

$$(18) \quad \Lambda_{j,i}^v \equiv \frac{\Omega_{j,i}(1 - \Omega_{j,i})}{E^c[\Omega_{j,i}(1 - \Omega_{j,i})]} \Lambda_i^c,$$

the next proposition characterizes the two elasticities of aggregate factor demand in terms of the corresponding firm-level elasticities.

PROPOSITION 1: *Let $E^x[\cdot]$, $\text{var}^x(\cdot)$, and $\text{cov}^x(\cdot, \cdot)$ denote the expectation, variance, and covariance operators with respect to one of the four different firm-level distributions Λ_i^x with $x \in \{r, p, c, v\}$ defined by equations (13), (15), (17), and (18). The aggregate elasticity of substitution and the aggregate output elasticity of relative factor demand for factor j satisfy*

$$(19) \quad \bar{\sigma}_j = (1 - \nu_j) E^v[\sigma_{j,i}] + \nu_j \lambda \frac{\bar{\mathcal{E}}}{1 + \lambda \bar{\mathcal{E}}^m} \frac{\text{var}^p(\Omega_{j,i}^m)}{\text{var}^c(\Omega_{j,i})},$$

$$(20) \quad \bar{\eta}_j = (1 - \nu_j) E^v[\eta_{j,i}] - \nu_j \lambda \frac{\bar{\mathcal{E}}}{1 + \lambda \bar{\mathcal{E}}^m} \left[\frac{\text{cov}^p(\Omega_{j,i}^m, \mathcal{E}_i^m)}{\text{cov}^c(\Omega_{j,i}, \mathcal{E}_i)} - \frac{1 + \lambda \bar{\mathcal{E}}^m}{\bar{\mathcal{E}}} \frac{1}{\lambda} \right] \\ \times \frac{\text{cov}^c(\Omega_{j,i}, \mathcal{E}_i)}{\text{var}^c(\Omega_{j,i})},$$

²³ The comparative statics of the allocation of output across firms with respect to factor prices and to the scale of aggregate output are presented in Appendix A.B.

where $\nu_j \equiv \text{var}^c(\Omega_{j,i})/\bar{\Omega}_j(1 - \bar{\Omega}_j) \in [0, 1]$ is the normalized variance of factor intensities across firms under the cost distribution Λ_i^c .

PROOF:

See Section OA.1 of the online Appendix.

Proposition 1 decomposes the aggregate elasticities of demand to within-firm and between-firm components. It thus generalizes the standard results for the aggregate elasticity of substitution under the assumption of CRS firm-level technologies (e.g., Oberfield and Raval 2021) to general firm-level technologies. Let us first consider a scale-invariant case ($\eta_{j,i} = 0$ and $\Omega_{j,i}^m = \Omega_{j,i}$) where all firms have a homogeneous and constant degree of returns to scale ($\mathcal{E}_i = \mathcal{E}$ and $\mathcal{E}_i^m = \mathcal{E} - 1$). In this case, equations (14)–(17) imply $\bar{\mathcal{E}}^m = \bar{\mathcal{E}} - 1 = \mathcal{E} - 1$ and $\Lambda_i^p \equiv \Lambda_i^c$. Therefore, applying the proposition leads to the following expression for the aggregate elasticity of substitution:

$$(21) \quad \bar{\sigma}_j = (1 - \nu_j)E^v[\sigma_{j,i}] + \nu_j \frac{\mathcal{E}}{1 + \lambda(\mathcal{E} - 1)}\lambda,$$

and the aggregate production function is scale invariant, i.e., $\bar{\eta}_j = 0$. Equation (21) shows that, so long as the degree of returns to scale is sufficiently low to allow for an equilibrium to exist ($\mathcal{E} > 1 - 1/\lambda$), a higher degree of returns to scale (lower $\mathcal{E}$) leads to a stronger contribution from between-firm reallocations and thus a larger aggregate elasticity of substitution. In the CRS case ($\mathcal{E} = 1$), the aggregate elasticity of substitution is a convex combination of the weighted mean of firm-level elasticity of substitution $E^v[\sigma_{j,i}]$ and the elasticity of demand λ .

Allowing for heterogeneity in the degrees of returns to scale $\mathcal{E}_i$ across firms, while still maintaining firm-level scale invariance ($\eta_{j,i} = 0$ and $\Omega_{j,i}^m = \Omega_{j,i}$) introduces new channels that shape the elasticities of aggregate factor demand. In particular, when the demand elasticity λ is greater than unity, the pass-through distribution $\Lambda_i^p \propto \{1/[\mathcal{E}_i - (1 - 1/\lambda)]\}\Lambda_i^r$ puts higher weights on firms with higher returns to scale compared to the cost distribution $\Lambda_i^c \propto (1/\mathcal{E}_i)\Lambda_i^r$. Thus, equation (16) implies that the aggregate cost elasticities satisfy $\bar{\mathcal{E}}^m < \bar{\mathcal{E}} - 1$. Consequently, equation (19) suggests that this case may lead to a stronger reallocation effect compared to the homogeneous case with the same level of aggregate returns to scale. Moreover, in equation (19) we have to now additionally account for the ratio of the variance of IT intensities under the pass-through distribution ($\text{var}^p(\Omega_{j,i})$) to that under the cost distribution ($\text{var}^c(\Omega_{j,i})$).²⁴

Most interestingly, in this case equation (20) shows that the aggregate technology can feature scale dependence despite the firm-level scale invariance. With heterogeneity in returns to scale, a rise in aggregate output reallocates production away from firms with high cost elasticities. To the extent that high-IT intensity firms (weighted by their respective pass-throughs) have lower cost elasticities, this reallocation leads

²⁴Note that in the special case where the elasticity of substitution across firm outputs is unity ($\lambda = 1$), the pass-through distribution defined in equation (17) coincides with the cost distribution in equation (15), and we find a result similar to the CRS case: $\bar{\sigma}_j = (1 - \nu_j)E^v[\sigma_{j,i}] + \nu_j$ and $\bar{\eta}_j = 0$.

to a positive aggregate output elasticity of relative IT demand $\bar{\eta}_j > 0$. The opposite is true if high-IT intensity firms have higher cost elasticities. Appendix A.C uses this insight to provide a microfoundation for nonhomothetic technology at the firm level.

Finally, allowing for scale dependence in firm-level technology, we have to additionally account in equations (19) and (20) for the fact that the elasticities of marginal cost $(\Omega_{j,i}^m, \mathcal{E}_i^m)$ deviate from those of cost $(\Omega_{j,i}, \mathcal{E}_i)$. If the output elasticity of relative IT demand is positive ($\eta_{j,i} > 0$), equation (9) suggests that the pass-through exceeds the factor intensity, and equation (19) implies that the contribution of reallocation to the aggregate elasticity of substitution may become larger than the case with scale-invariant firm-level technologies. Equation (20) shows that the within-firm component in the aggregate output elasticity of relative factor demand $\bar{\eta}_j$ reflects the average degree of scale dependence in firm-level technology $E^v[\eta_{j,i}]$. The between-firm component now reflects the covariances between pass-through and marginal cost elasticity $\text{cov}^p(\Omega_{j,i}^m, \mathcal{E}_i^m)$ and the cost share and cost elasticity $\text{cov}^c(\Omega_{j,i}, \mathcal{E}_i)$. In our empirical application of these results in Section IVA, we will see that the between-firm effects may undo the within-firm effects, leading to scale invariance in the aggregate despite the presence of firm-level scale dependence.

General Equilibrium Response to a Factor Price Shock.—We are now prepared to study the response of the economy to a fall in the relative price of IT. For the purpose of this exercise, we assume that the aggregate supplies for $\bar{\mathbf{X}}$ of all factors are exogenous and perfectly inelastic. We then study a shock to the price of a given factor $j \equiv I$ of production, in our case IT. This shock is in turn driven by an underlying shock to the (effective) aggregate supply $\bar{X}_I$ of the factor, keeping the supplies of all other factors fixed.

First, let us consider the (partial equilibrium) comparative statics of our aggregate variables of interest with respect to the IT prices. Applying Lemma 1 to the aggregate cost and demand functions, we find the macro parallels to firm-level equations (9) and (10):

$$(22) \quad \frac{\partial \log \bar{P}}{\partial \log W_I} = \bar{\Omega}_I^m = \bar{\Omega}_I \left(1 + \frac{\bar{\Omega}_N}{\bar{\mathcal{E}}} \bar{\eta}_I \right),$$

$$(23) \quad \frac{\partial \log F\bar{S}_N}{\partial \log W_I} = \bar{\Omega}_I \left(\bar{\sigma}_I - \frac{\bar{\Omega}_N}{\bar{\mathcal{E}}} \bar{\eta}_I - 1 \right).$$

Equation (22) shows that, when there is aggregate scale dependence in IT demand $\bar{\eta}_I \neq 0$, the effect of a shock to the price of IT on the aggregate price index rises approximately by a factor $\bar{\eta}_I$ due to the endogenous response of returns to scale. Equation (23) shows that such aggregate scale dependence also lowers the effective elasticity of substitution in measuring the factor distribution of income by approximately $\bar{\eta}_I$.

In general equilibrium, aggregate output and other factor prices are likely to respond to falling IT prices. To simplify the analysis, we next impose a separability restriction on firm-level technology, assuming

$$(24) \quad \mathcal{F}_i(\mathbf{X}_i) = \mathcal{G}_i(X_{I,i}, X_{N,i}), \quad X_{N,i} \equiv \mathcal{X}_N(\mathbf{X}_{-I,i}),$$

where $\mathcal{X}_N(\cdot)$ is a CRS aggregator of the vector $\mathbf{X}_{-I,i}$ of non-IT inputs. Since we assume identical patterns of substitutability between IT and all non-IT inputs, a shock to the aggregate supply (or productivity) of IT inputs shifts the price of IT relative to that of the bundle of non-IT inputs, leaving all other relative prices intact. Moreover, based on this assumption, the effect of such a shock on the share of all non-IT factors $\bar{FS}_j$ for $j \neq I$ is proportional to the relative change in the share of non-IT factors $\bar{FS}_N$. Hence, we can characterize the general equilibrium response using only the same aggregate technological elasticities derived in Section IIB, accounting both for the direct effect of the factor price change and the indirect effect through the resulting response of the aggregate output.

PROPOSITION 2: *Assume firm-level technology satisfies condition (24), and consider a shock to the relative price of IT inputs $W \equiv W_I/W_N$ that leaves the firm-level technologies and the aggregate supply of all non-IT factors unchanged. To the first order of approximation in the change in the IT price shock, the resulting impact on the aggregate output and on the income share of non-IT inputs are given by*

$$(25) \quad \frac{d \log \bar{Y}}{d \log W} = -\bar{\Omega}_I \frac{\bar{\sigma}_I}{\bar{\mathcal{E}} - \bar{\Omega}_I \bar{\eta}_I},$$

$$(26) \quad \frac{d \log \bar{FS}_N}{d \log W} = \bar{\Omega}_I \left[\bar{\sigma} \left(\frac{1 + \bar{\mathcal{E}}^m}{\bar{\mathcal{E}} - \bar{\Omega}_I \bar{\eta}_I} \right) - \frac{\bar{\Omega}_N}{\bar{\mathcal{E}}} \bar{\eta}_I - 1 \right].$$

PROOF:

See Section OA.1 of the online Appendix.

To understand the insights contained in equations (25) and (26), let us consider the benchmark with CRS firm-level production functions, in which case we have $\bar{\mathcal{E}} = 1$ and $\bar{\mathcal{E}}^m = \bar{\eta}_I = 0$. In this case, the impact on the aggregate output is simply given by the product of the share of IT in costs and the aggregate elasticity of substitution $\bar{\Omega}_I \cdot \bar{\sigma}$. The general equilibrium response of the income share of the non-IT factor is simply given by $\bar{\Omega}_I(\bar{\sigma}_I - 1)$. In line with the standard results (e.g., Karabarbounis and Neiman 2014; Oberfield and Raval 2021), a fall in IT prices lowers the share of non-IT inputs in the aggregate income only if the aggregate elasticity of substitution $\bar{\sigma}$ exceeds unity.

More generally, equation (25) shows that the effect of the fall in IT prices on the aggregate output additionally depends on the aggregate cost elasticity $\bar{\mathcal{E}}$ and the aggregate output elasticity of relative IT demand $\bar{\eta}_I$. In particular, if the aggregate IT demand rises in output ($\bar{\eta}_I > 0$), the fall in IT prices leads to a stronger impact on the aggregate output compared to the scale-invariant benchmark. This aggregate output response further appears in the general equilibrium response of the factor distribution of income in equation (26), which can be compared with the partial equilibrium response in equation (23).

We can similarly characterize the general equilibrium responses of other aggregate variables of interest using the response of aggregate output characterized above. For instance, consider the aggregate profit share, which from equation (5) satisfies $\bar{\Pi}/(\bar{P}\bar{Y}) \equiv (\bar{P}\bar{Y} - \bar{C})/(\bar{P}\bar{Y}) = 1 - 1/(\mu\bar{\mathcal{E}})$. We can now derive the response of

this share through the general equilibrium response of the aggregate cost elasticity, given by

$$(27) \quad \frac{d}{d \log W} \left(\frac{\bar{\Pi}}{\bar{P} \bar{Y}} \right) = \frac{1}{\mu \bar{\mathcal{E}}} \left\{ \frac{\bar{\Omega}_N}{\bar{\mathcal{E}}} \bar{\Omega}_I \bar{\eta}_I + [\bar{\mathcal{E}}^m - (\bar{\mathcal{E}} - 1)] \frac{d \log \bar{Y}}{d \log W} \right\}.$$

Falling IT prices impact the profit share either directly through the returns to scale channel, the first term in equation (27), or indirectly through the aggregate cost superelasticity, the term involving the response of aggregate output in equation (27).

Finally, to study the response in the concentration of production among the largest firms, we derive the general equilibrium response of firm-level market shares to the changes in the price of IT. Lemma 3 and equation (13) together imply

$$(28) \quad \frac{d \log \Lambda_i^r}{d \log W} = - \frac{\lambda - 1}{1 + \lambda \mathcal{E}_i^m} \left[\Omega_{I,i}^m - \bar{\Omega}_I^m + (\mathcal{E}_i^m - \bar{\mathcal{E}}^m) \frac{d \log \bar{Y}}{d \log W} \right].$$

As we saw in equation (25), the response of aggregate output is negative $d \log \bar{Y} / d \log W_I < 0$. Thus, when the price of IT falls, the response of the market share grows in the pass-through $\Omega_{I,i}^m$ and falls in the marginal cost elasticity $\mathcal{E}_i^m$. Since the pass-through is likely to rise in firm size due to the scale dependence in IT demand, the first channel predicts that falling IT prices lead to a potential rise in concentration, as proxied by the market share of the top quantiles of firm size.

In Section IVB, we will further revisit these results and their quantitative magnitude in the case of the French economy using our firm-level data.

III. Estimation: Scale-Dependent Production Function

Assuming smooth production functions, our theory in Section II showed how two elasticities of the firm-level IT factor demand, the elasticity of substitution $\sigma_{I,i}$ (accounting for the substitution channel) and the output elasticity of relative demand $\eta_{I,i}$ (accounting for the returns to scale channel), together shape the technological response to a fall in IT prices. To take the predictions of our theory to the data, we need to estimate these key technological elasticities at the firm level. In this section, we tackle this estimation problem. First, in Section IIIA, we introduce a parametric family of production functions that allows us to capture not only substitution between IT and non-IT inputs but also potential scale dependence in IT demand. Using this specification, we then lay out in Section IIIB our approach to the identification of the production function and continue to present the estimation results in Section IIIC.

A. Parameterizing Firm Technology: Nonhomothetic CES Production

Common specifications of firm-level technology in models involving firm heterogeneity, e.g., Cobb-Douglas or CES, typically assume scale dependence $\eta_{I,i} \equiv 0$. We consider a generalization of these specifications that maintains the assumption of constant elasticity of substitution ($\sigma_{I,i} \equiv \sigma$), while extending it to additionally allow for a constant output elasticity of relative IT demand ($\eta_{I,i} \equiv \eta$). The latter

assumption is in line with the evidence in Figure 1 for a stable relationship between IT intensity and firm size.²⁵ Since our specification nests Cobb-Douglas and CES specifications, it also allows us to transparently examine the implications of scale dependence in IT demand.

We assume a nested firm-level production technology that features heterogeneity across firms and over time. At the lower level of the nest, the vector of factor inputs $\mathbf{X}_{it}$ of firm i at time t is aggregated into two bundles of IT inputs $X_{I,it}$ and non-IT inputs $X_{N,it}$ following

$$(29) \quad X_{N,it} \equiv K_{it}^{\alpha} L_{it}^{1-\alpha}, \quad X_{I,it} \equiv S_{it}^{\beta} H_{it}^{1-\beta},$$

where K_{it} , L_{it} , S_{it} , and H_{it} stand for non-IT capital, labor, software, and hardware inputs. We assume that the upper level of the nest belongs to the class of *nonhomothetic CES production functions* (Sato 1974, 1977; Hanoch 1975b; Comin, Lashkari, and Mestieri 2021), such that

$$(30) \quad Y_{it} = \mathcal{F}_{it}(\mathbf{X}_{it}) \equiv nhCES(e^{\theta_{it}} X_{N,it}, e^{\theta_{it} + \phi_{it}} X_{I,it}; \sigma, \gamma, \epsilon),$$

where θ_{it} and ϕ_{it} are the factor-symmetric and IT-biased (log) productivity states, which account for the variations in technology across firms and over time. The nhCES production function $Y = nhCES(X_N, X_I)$ in equation (30) is implicitly defined through the constraint

$$(31) \quad \left(\frac{X_N}{Y^{\gamma}} \right)^{\frac{\sigma-1}{\sigma}} + \left(\frac{X_I}{Y^{\gamma+\epsilon}} \right)^{\frac{\sigma-1}{\sigma}} = 1.$$

We assume that parameters (γ, σ) are positive valued and satisfy $\epsilon > -\gamma$ to ensure the production function is globally monotonically increasing in both inputs. The nhCES family is uniquely defined by assuming that both the substitution elasticity and the output elasticity of relative demand are constant (Comin, Lashkari, and Mestieri 2021).

Motivation for nhCES Technology.—We can provide a construction of the production function in equation (31) that offers some intuitions about the potential underlying mechanisms giving rise to scale dependence in IT demand. Let us follow Lucas (1978) in distinguishing organization as a distinct input to firm production. Assume that the technology is characterized by an organizational input O that is combined with the bundle of non-IT inputs to form output Y according to $Y = \left(O^{(\sigma-1)/\sigma} + X_N^{(\sigma-1)/\sigma} \right)^{\sigma/\gamma(\sigma-1)}$, where the parameter γ controls the degree of returns to scale. The key assumption is that the organizational input of the firm endogenously varies according to $O = X_I \times Y^{-\epsilon}$, implying that firms (i) face diminishing returns to the organizational input as the output Y rises due

²⁵ Appendix B.B provides more direct evidence on the stability of the relation between size and IT intensity across different brackets of size. We provide a detailed discussion of an alternative production function with fixed costs of IT adoption that leads to a positive but variable output elasticity of relative IT demand $\eta_{I,i}$ in Section OA.4.1 of the online Appendix.

to potential increase in organizational complexity and (ii) can raise the organizational input by deploying IT inputs X_I . The two equations together lead to equation (31), providing a specific interpretation for the resulting scale dependence: larger firms optimally choose higher IT intensities in response to the fall in their organizational efficiency due to higher complexities of production. We will return to and expand on this motivation for scale dependence in IT demand in Section VA.

nhCES Cost Function and Factor Demand.—Following Assumption 1, we can apply cost minimization. Solving this problem for the nhCES production function leads to the following cost function:²⁶

$$(32) \quad C_{it} = C_{it}(Y_{it}; W_t) = e^{-\theta_{it}} Y_{it}^\gamma \left[1 + (e^{-\phi_{it}} W_t Y_{it}^\epsilon)^{1-\sigma} \right]^{\frac{1}{1-\sigma}},$$

where we have defined the relative price of IT $W_t \equiv W_{I,t}/W_{N,t}$, setting the price of the bundle of non-IT inputs in every period as the numeraire. The relative IT demand and the cost share of IT satisfy

$$(33) \quad \frac{W_t X_{I,it}}{X_{N,it}} = (e^{-\phi_{it}} W_t Y_{it}^\epsilon)^{1-\sigma}, \quad \Omega_{I,it} = \frac{(e^{-\phi_{it}} W_t Y_{it}^\epsilon)^{1-\sigma}}{1 + (e^{-\phi_{it}} W_t Y_{it}^\epsilon)^{1-\sigma}}.$$

The expressions for relative IT demand in equation (33) demonstrate the defining characteristic of the nhCES production function: a log-linear relationship between the relative IT demand, the relative price of IT, the scale of firm outputs, and IT-biased productivity. As a result, as we already suggested above, our two main elasticities of IT demand defined in equations (6) and (7) take constant values across all firms, that is,

$$(34) \quad \sigma_{I,it} = \sigma, \quad \eta_{I,it} = \eta = (1 - \sigma)\epsilon.$$

B. Estimation Strategy

In order to estimate the nhCES production function, let us first express it in logarithmic terms. Using lowercase letters to denote the logarithm of the corresponding variable, e.g., $y_{it} \equiv \log Y_{it}$ and $x_{I,it} \equiv \log X_{I,it}$, equation (31) implies

$$(35) \quad y_{it} = \frac{1}{\gamma} \frac{\sigma}{\sigma - 1} \log \left[e^{\frac{\sigma-1}{\sigma} x_{N,it}} + e^{\frac{\sigma-1}{\sigma} (x_{I,it} + \phi_{it} - \epsilon y_{it})} \right] + \frac{1}{\gamma} \theta_{it},$$

where we have $x_{N,it} \equiv \alpha k_{it} + (1 - \alpha) l_{it}$ and $x_{I,it} \equiv \beta s_{it} + (1 - \beta) h_{it}$ from equation (29). Equation (35) displays one major distinction compared to the common specifications such as Cobb-Douglas or Translog typically estimated in the literature. Besides the factor-symmetric productivity state θ_{it} common in the latter specifications, our setting additionally allows for unobserved IT-augmenting productivity state ϕ_{it} .

²⁶For the derivations of equations (32) and (33), see Section OA.1 of the online Appendix.

To better appreciate the importance of including IT-augmenting productivity state in our setting, let us also rewrite in logarithmic terms the IT demand equation (33), implied by our benchmark Assumption 1 on factor markets, as

$$(36) \quad x_{I,it} - x_{N,it} = -\sigma w_t + (1 - \sigma)\epsilon y_{it} + (\sigma - 1)\phi_{it}.$$

Equation (36) highlights two important roles of the unobserved productivity ϕ_{it} . First, it offers an alternative to firm scale y_{it} (and log relative IT prices w_t) as a source of variations in IT intensity across firms and over time. Second, it highlights the key challenge for identifying the elasticity of substitution σ since we cannot recover it with a simple OLS regression due to potential covariance between log relative IT price w_t and the unobserved productivity ϕ_{it} .

In order to extend the production function estimation strategy to our setting, which involves sources of unobserved heterogeneity θ_{it} and ϕ_{it} , we take the following strategy. Since Assumption 1, which leads to the demand equation (36), may be too strong to be taken to the micro data, we first partially relax this assumption to derive a generalized demand equation. This step allows us to derive a corresponding expression for ϕ_{it} in terms of observables and model parameters and write the vector of productivities $\tilde{\theta}_{it} \equiv (\theta_{it}, \phi_{it})'$ as a function of observed data and model parameters. Next, we generalize the standard identification assumptions in the production function estimation to our two-dimensional productivity space, which allows us to estimate the model parameters.²⁷

IT-Biased Productivity.—To derive equation (36), we relied on Assumption 1, which states that firms face competitive factor markets and that they can “flexibly” adjust all these inputs without adjustment costs. While these assumptions provide a reasonable benchmark for analyzing the long-run behavior of firms in our aggregative model, they may not be tenable when applied to the year-to-year changes in the observed capital input choices of firms in our data. Thus, for the purpose of estimating the production function, we partially relax Assumption 1 about the factor markets as follows.

ASSUMPTION 1' (Generalized Setting for Factor Input Decisions/Markets): *Factor input decisions and markets satisfy the following conditions.*

- (i) *Hardware and non-IT capital may involve adjustment costs and other firm-level frictions.*
- (ii) *Labor is a flexible input with perfectly competitive local markets and wages.*
- (iii) *Software is a flexible input with a perfectly competitive national market and price.*

²⁷ For a recent example of this approach applied to the estimation of homothetic CES production functions, see Doraszelski and Jaumandreu (2018).

The first part of the generalized assumption allows for potential adjustment costs and other potential firm-level frictions in investments in hardware and non-IT capital, e.g., those stemming from financial constraints. Note that the frictionless setting for these two inputs considered under the benchmark Assumption 1 is still a special case of Assumption 1'. The second part of the generalized assumption assumes that firms operate in perfectly competitive local labor markets and allows for variations in firm-level wages in the cross section of locations. In practice, depending on the location n of the headquarters of a given firm i , we use a measure of average wage at the corresponding local employment area to construct a location-specific wage $W_{L,nt}$.²⁸ Again, note that the case with a single, national wage considered under the benchmark Assumption 1 is a special case of Assumption 1'. Software remains the only factor following our benchmark Assumption 1 and features a frictionless and national market with a price $W_{S,t}$.²⁹

The following lemma generalizes equation (36) to derive a relationship between the IT-biased productivity ϕ_{it} , the observed data, and the parameters of the production function under Assumption 1'.³⁰

LEMMA 2: *Under Assumption 1' and under general forms of adjustment costs for non-IT capital and hardware, the firm's choices of inputs in each period satisfy*

$$(37) \quad s_{it} - l_{it} = -\sigma(w_{S,t} - w_{L,nt}) + (1 - \sigma)\epsilon y_{it} \\ + (1 - \sigma)[\alpha(k_{it} - l_{it}) - (1 - \beta)(h_{it} - s_{it})] + (\sigma - 1)\phi_{it},$$

where $w_{S,t} - w_{L,nt}$ is the (location-specific) logarithm of the price of software relative to wages.

PROOF:

See Section OA.1 of the online Appendix.

Since Assumption 1' maintains two flexible inputs, labor in the non-IT and software in the IT bundle, the relative prices of the two inputs and their respective marginal products are equalized. The implied cost minimization equation (37) modifies equation (36) in three different ways. First, it focuses on the logarithm of the

²⁸ See Section OA.2.5.2 of the online Appendix for further details about the construction of the local wage.

²⁹ This assumption implies that firms can disinvest in software if they receive large negative productivity shocks. Note that there is scant evidence on potential asymmetries between purchase and liquidation value of software inputs. Even if firms face disinvestment frictions, there are reasons to believe that states in which firms need to disinvest in software are unlikely. First, since software has a high depreciation rate (31 percent based on the series provided by the INSEE), firms can substantially reduce their software stock without disinvestment. Second, the strong negative trend in the user cost of software implies that we need fairly large negative productivity shocks to make disinvestment optimal. Third, large negative productivity shocks are unlikely since typical estimates suggest persistent firm-level productivity processes. Fourth, since we allow for adjustment costs in non-IT and hardware capital, a downsizing firm that faces binding frictions on those two inputs would not choose to substantially lower its labor and software inputs due to complementarity in production.

³⁰ The demand equation (37) is a strict generalization of equation (36) in the following sense. Equation (36) does not follow solely from the relaxed Assumption 1' since it requires first-order conditions with respect to non-IT capital and hardware, which may not hold under this assumption. In contrast, equation (37) indeed follows from the benchmark Assumption 1 since it only requires first-order conditions with respect to software and labor, which hold under this assumption.

ratio of the flexible inputs, the software-to-labor ratio, as opposed to the ratio of IT to non-IT inputs. Second, it allows firms to face different (log) prices of software relative to wages $w_{S,t} - w_{L,nt}$, e.g., if they are located in regions with different wages. Depending on relative price of software to labor, they may substitute toward or away from software with elasticity σ . Third, firms with different (log) capital-to-labor ratios $k_{it} - l_{it}$ and hardware-to-software ratios $h_{it} - s_{it}$ may have different software-to-labor ratios.³¹

Lemma 2 generalizes the cost minimization equation (36) to accommodate potential firm-level distortions in hardware and non-IT capital input choices. When all factors are frictionless, capital-to-labor and hardware-to-software ratios are equalized across firms within a given location.³² In the presence of distortions, the two ratios vary across firms even within a given location. The variations in these ratios account for distortions across firms in the shadow relative prices of capital relative to labor and hardware relative to software, respectively. This is why the effect of these ratios in equation (37) on the (log) ratio labor intensity of software $s_{it} - l_{it}$ also depends on the elasticity of substitution σ .³³

Equation (37) allows us to write the IT-biased productivity ϕ_{it} in terms of the observables and model parameters. Substituting this expression for ϕ_{it} in equation (35) then allows us to recover log output y_{it} in terms of observables, model parameters, and factor-symmetric productivity θ_{it} . Let $\mathbf{d}_{it} \equiv (w_{S,t} - w_{L,nt}, l_{it}, k_{it}, s_{it}, h_{it}, y_{it})$ denote the vector of all relevant data observations for firm $i \in \mathcal{J}_{nt}$ in location n at time t . Letting $\mathbf{y}_{it} \equiv (y_{it}, s_{it})'$, equations (35) and (37) together define a vector function $\mathbf{f}(\mathbf{d}_{it}; \varsigma) \equiv (f_y(\mathbf{d}_{it}; \varsigma), f_s(\mathbf{d}_{it}; \varsigma))'$ such that the additive residual is a vector of scaled productivity states $\tilde{\boldsymbol{\vartheta}}_{it} \equiv (\theta_{it}/\gamma, (\sigma - 1)\phi_{it})'$, which satisfies³⁴

$$(38) \quad \mathbf{y}_{it} = \mathbf{f}(\mathbf{d}_{it}; \varsigma) + \tilde{\boldsymbol{\vartheta}}_{it}.$$

Next, we discuss the identification assumptions that allow us to estimate equation (38).

Identification.—The main identification challenge in estimating the production function based on equation (38) is the potential correlations between the contemporaneous productivity $\tilde{\boldsymbol{\vartheta}}_{it}$ and firm-level decisions $\mathbf{d}_{it}$. In line with the assumptions of the dynamic panel literature (e.g., Blundell and Bond 2000; Akerberg, Caves, and Frazer 2015; Akerberg 2016), we assume an AR(1) process for the evolution of firm-level productivity given by $\tilde{\boldsymbol{\vartheta}}_{it} = \boldsymbol{\mu}_t(\tilde{\boldsymbol{\vartheta}}_{it-1}; \rho) + \mathbf{u}_{it}$, where productivity innovations

³¹Note that this result also poses a challenge for potential alternative reduced-form identification strategies relying solely on demand shocks as potential instruments for output y_{it} in equation (37). Since output y_{it} may have an impact on capital-to-labor and hardware-to-software ratios, due potentially to adjustment costs or financial constraints, equation (37) shows that we additionally need instruments for the latter two ratios to identify ϵ . See Section OA.5.7 of the online Appendix for more details.

³²Note that under Assumption 1, Cobb-Douglas aggregators of the two bundles of non-IT and IT inputs imply that the optimal relative capital-to-labor and hardware-to-software ratios satisfy $k_{it} - l_{it} = \log[(\alpha)/(1 - \alpha)] + w_{L,nt} - w_{K,t}$ and $h_{it} - s_{it} = \log[(1 - \beta)/\beta] + w_{S,t} - w_{H,t}$ and are constant across firms within a given location.

³³One key consequence of this point is that the variations in capital-to-labor and hardware-to-software ratios help in the identification of the elasticity of substitution σ based on equation (37). As we will see below, the variations in these ratios are instrumented by their corresponding lagged values in a model-consistent way.

³⁴See the derivation of equation (38) and the explicit expressions for $\mathbf{f}(\cdot; \varsigma)$ and its components $f_y(\cdot; \varsigma), f_s(\cdot; \varsigma)$ in Section OA.1.2 of the online Appendix.

$\mathbf{u}_{it} \equiv (u_{\theta,it}, u_{\phi,it})'$ are zero mean and normally distributed. The conditional expectation of the productivity states are given by

$$(39) \quad \mu_t(\boldsymbol{\vartheta}_{it-1}; \rho) \equiv \begin{pmatrix} \rho_{\theta\theta} & \rho_{\theta\phi} \\ \rho_{\phi\theta} & \rho_{\phi\phi} \end{pmatrix} \boldsymbol{\vartheta}_{it-1} + \begin{pmatrix} \eta_\theta + \mu_\theta t \\ \eta_\phi + \mu_\phi t \end{pmatrix}.$$

The vector $(\mu_\theta, \mu_\phi)'$ captures the trend in mean industry productivity over time, and the tuple $\rho \equiv (\rho_{\theta\theta}, \rho_{\theta\phi}, \rho_{\phi\theta}, \rho_{\phi\phi}, \eta_\theta, \eta_\phi, \mu_\theta, \mu_\phi)$ includes all parameters of the Markov process.

The dynamics of productivity states leads to a Markov structure on the residuals in equation (38):

$$(40) \quad \tilde{\boldsymbol{\vartheta}}_{it} = \tilde{\mu}_t(\tilde{\boldsymbol{\vartheta}}_{it-1}; \tilde{\rho}) + \tilde{\mathbf{u}}_{it},$$

where $\tilde{\mathbf{u}}_{it}$ denotes a contemporaneous productivity innovation, the function $\tilde{\mu}_t$ characterizes the persistence in productivity, and $\tilde{\rho} \equiv (\boldsymbol{\varsigma}, \rho)$.³⁵

Following the dynamic panel approach, we then assume that when firms make decisions at time $t - 1$, they do not know the future productivity innovations $\tilde{\mathbf{u}}_{it}$ in equation (40). In other words, lagged firm choices $\mathbf{d}_{it-1}$ are orthogonal to current productivity innovations $\tilde{\mathbf{u}}_{it}$. Thus, we can use these lagged values, e.g., lagged labor l_{it-1} , as instruments for current firm decisions, e.g., labor l_{it} , in equation (38), while controlling for the persistence in productivity shocks.³⁶

This approach provides us with instruments for all but one component of the vector of firm outcomes $\mathbf{d}_{it}$, i.e., the log price of software relative to wages $w_{S,t} - w_{L,nt}$. To construct a valid instrument for this variable, we borrow from Oberfield and Raval (2021) and construct shift-share instruments z_{nt} for log local wage $w_{L,nt}$ faced by firms in each location n . This instrument is given by interacting the initial industrial composition of each location with the time evolution of each industry's wage bill at the national level.³⁷ We expect wages to rise relatively more in locations in which employment was initially concentrated in those industries that witness higher demand shocks at the national level. Together, we can summarize all our identification assumptions as follows.

ASSUMPTION 2 (Identification Assumptions): *Let $\mathcal{I}_{it}$ denote the information set of the firm i at time t , which includes all observables up to time t , including the*

³⁵ See the derivation of equation (40) and the explicit expressions for $\tilde{\mu}_t(\cdot; \rho, \boldsymbol{\varsigma})$ in Section OA.1.2 of the online Appendix.

³⁶ Benchmark approaches to production function estimation typically involve an additional step to construct proxy variables for productivity $\tilde{\boldsymbol{\vartheta}}_{it}$ (e.g., Olley and Pakes 1996; Levinsohn and Petrin 2003). The reason is that these approaches assume another source of error e_{it} in the residuals of equation (38) in the form $\mathbf{y}_{it} = \mathbf{f}(\mathbf{d}_{it}; \boldsymbol{\varsigma}) + \tilde{\boldsymbol{\vartheta}}_{it} + e_{it}$. Here, we simplify the problem by assuming that all unobserved errors are captured in the productivity variable $\tilde{\boldsymbol{\vartheta}}_{it}$ that is known to the firm i at time t .

³⁷ See Section OA.2.9.2 of the online Appendix for more details. Following Bartik (1991), a vast empirical literature has relied on shift-share instruments such as that used here to construct exogenous variations in wages at the regional level. For recent contributions to the analysis and design of these instruments in settings involving micro data, see Adao, Kolesár, and Morales (2019); Borusyak, Hull, and Jaravel (2022); Goldsmith-Pinkham, Sorkin, and Swift (2018).

vector of productivities $\mathbf{\Theta}_{it}$. We assume that all firm decisions at time t depend on its information set $\mathcal{I}_{it}$ and

(i) The evolution of the productivity states satisfies equation (39) and $E[\mathbf{u}_{it} | \mathcal{I}_{it-1}] = \mathbf{0}$.

(ii) For firms in location n , IT-biased productivity innovations satisfy $E[u_{\phi,it} | z_{nt}] = 0$.

Relying on Assumption 2, we now construct the moment conditions that allow us to estimate the parameters of the production function and the Markov process. Let $\tilde{\mu}_{\theta,t}$ and $\tilde{\mu}_{\phi,t}$ stand for the components of the conditional expectation of the scaled productivity $\tilde{\boldsymbol{\mu}}_t \equiv (\tilde{\mu}_{\theta,t}, \tilde{\mu}_{\phi,t})'$. We consider the following moment conditions:

$$(41) \quad E\left[\left(y_{it} - f_y(\mathbf{d}_{it}; \boldsymbol{\varsigma}) - \tilde{\mu}_{\theta,t}(\mathbf{y}_{it-1} - \mathbf{f}(\mathbf{d}_{it-1}; \boldsymbol{\varsigma}); \rho, \boldsymbol{\varsigma})\right) \times z_{it}^y\right] = 0,$$

$$(42) \quad E\left[\left(s_{it} - f_s(\mathbf{d}_{it}; \boldsymbol{\varsigma}) - \tilde{\mu}_{\phi,t}(\mathbf{y}_{it-1} - \mathbf{f}(\mathbf{d}_{it-1}; \boldsymbol{\varsigma}); \rho, \boldsymbol{\varsigma})\right) \times z_{it}^s\right] = 0,$$

where we have eight instruments z_{it}^y for the log output and eight instruments z_{it}^s for the log software stock:

$$(43) \quad z_{it}^y \in \{l_{it-1}, k_{it-1}, s_{it-1}, h_{it-1}, y_{it-1} - f_y(\mathbf{d}_{it-1}; \boldsymbol{\varsigma}), s_{it-1} - f_s(\mathbf{d}_{it-1}; \boldsymbol{\varsigma}), 1, t\},$$

$$(44) \quad z_{it}^s \in \{z_{nt}, k_{it-1} - l_{it-1}, s_{it-1} - h_{it-1}, y_{it-1} - f_y(\mathbf{d}_{it-1}; \boldsymbol{\varsigma}), \\ s_{it-1} - f_s(\mathbf{d}_{it-1}; \boldsymbol{\varsigma}), 1, t\},$$

where z_{nt} is the shift-share instrument for local wages in location n hosting firm i . We use the system of moment conditions (41) and (42) to estimate the production function in a nonlinear GMM framework.³⁸

C. Estimation Results

Our moment conditions relate the logarithm of inputs, the logarithm of output, and their respective lagged values. Therefore, we restrict the estimation to the unbalanced panel of firms for which all these values are observed in the data. Table 2 presents the estimated parameters of the production function for the pooled sample of all industries and for the sample of all manufacturing firms.³⁹

Let us begin with the scale dependence parameter ϵ , presented in the first row of Table 2. We find significant, sizable, and positive estimates for this parameter in the two samples of firms ($\epsilon = 0.43$ for all industries and $\epsilon = 0.47$ in manufacturing). We can reject the homotheticity of the production function. The second row of the

³⁸Section OA.5.2 of the online Appendix illustrates how different moments help identify the key model parameters, i.e., the scale dependence parameter ϵ and the elasticity of substitution σ in the structural estimation laid out above.

³⁹Table OA.13 in the online Appendix also reports the parameters of the Markov process.

TABLE 2—ESTIMATION RESULTS

		All industries	Manufacturing
Scale dependence parameter	ϵ	0.433 (0.026)	0.466 (0.027)
Elasticity of substitution	σ	0.280 (0.034)	0.131 (0.036)
Cost elasticity parameter	γ	0.939 (0.004)	0.946 (0.006)
Capital elasticity of non-IT	α	0.062 (0.003)	0.146 (0.007)
Software elasticity of IT	β	0.042 (0.034)	0.152 (0.034)
Observations	N	307,504	148,979

Notes: Results of the estimation procedure for the pooled sample of all firms in and for the pooled sample of manufacturing firms. Standard errors are reported in parentheses.

table presents the estimated elasticities of substitution. In both samples, we find values for the elasticity of substitution that are below unity ($\sigma = 0.28$ for all industries and $\sigma = 0.13$ in manufacturing), implying gross complementarity between IT and non-IT inputs.⁴⁰ In both samples, the estimated parameters ϵ and σ are in line with positive values for the output elasticity of relative IT demand and with gross complementarity between IT and non-IT inputs: $\epsilon > 0$ and $\sigma < 1$.

The third row of Table 2 presents our estimated values for the cost elasticity parameter γ . We find similar estimates for this parameter in both samples of firms ($\gamma \approx 0.94$). As we will see in equation (45), the cost elasticity and the IT intensity under the nhCES specification satisfy $\mathcal{E}_i = \gamma + \epsilon \Omega_{I,i}$. Since cost elasticity $\mathcal{E}_i$ has an inverse relationship with returns to scale and the IT intensity rises in firm size, the estimated parameters imply a negative relation between size and returns to scale. An estimated value for γ below unity implies increasing returns to scale for the very small firms, for which the IT intensity is negligible, $\Omega_{I,i} \approx 0$. In the sample of all industries, we find scale elasticities that move from 1.07 to 0.73 as we move from the smallest to the largest firms (the corresponding values are 1.06 to 0.71 for the sample of manufacturing firms). The smallest firms in the sample operate under increasing returns to scale, whereas the largest operate under decreasing returns to scale.

Table 2 also presents the values of the elasticities of non-IT capital α and software β , in their respective aggregators for the bundles of non-IT and IT inputs.⁴¹ These

⁴⁰ Oberfield and Raval (2021) and Doraszelski and Jaumandreu (2018) estimate the micro-level elasticity of substitution between capital and labor using identification strategies that account for potential factor-augmenting productivity shocks, and find values between 0.4 and 0.7. An earlier set of macroeconomic estimates find values slightly higher but still below unity (e.g., Antràs 2004; Klump, McAdam, and Willman 2007; Chirinko 2008). We also note that Karabarbounis and Neiman (2014) use an estimation strategy that relies on the cross-sectional variations in the industry-level data and finds a value above 1.

⁴¹ Note that the parameters α and β are distinct from the output elasticities of non-IT capital and software in the overall production function. For the nhCES production function, for instance, the output elasticity of labor is given by $(1 - \alpha)[(1 - \Omega_{I,i})/\mathcal{E}_i]$ and varies across firms. For the Cobb-Douglas production function, the output elasticity of labor is given as the product of $(1 - \alpha)$ and the constant output elasticity of the bundle of non-IT inputs (not shown in the table). Table OA.29 in the online Appendix reports the output elasticities of each of the four factors in the case of the Cobb-Douglas production function.

values imply lower output elasticities for non-IT capital and software compared to labor and hardware, respectively.

Section OA.5.1.1 of the online Appendix compares the results of estimating a CES production function (imposing $\epsilon = 0$) based on the same strategy. The Appendix also shows that our estimated nhCES technology accounts for the entire cross-sectional relationship between IT intensity and firm scale documented in Section IB through the effect of scale dependence.

D. Robustness

We also estimate parameters ϵ , σ , and γ across 22 industries at the A38 level of the aggregated NAF classification. The estimated values of the parameter ϵ are positive and significant for most of the industries, estimated values of the parameter σ are never above 1, and estimated values of the cost elasticity parameter γ are generally close to 1 across industries. The results are reported in Section OA.5.9 of the online Appendix (e.g., Figure OA.3).

We perform additional estimation exercises to examine the robustness of our estimates of the two key parameters of the model, the scale dependence parameter ϵ and the elasticity of substitution σ , to alternative choices of specification. Section OA.5.1 of the online Appendix presents three alternative specifications of the firm-level production function: (i) a gross output production function (as opposed to the value added production function considered here) in Section OA.5.1.2 of the online Appendix, (ii) a production function in which the IT bundle aggregate features complementarity between software and hardware (as opposed to the Cobb-Douglas aggregator considered here) in Section OA.5.1.3 of the online Appendix, and (iii) a production function with the IT bundle that includes an additional technical labor input in Section OA.5.1.4 of the online Appendix. In all cases, the estimates of our key parameters share the same broad patterns as in our benchmark specification: a positive and significant scale dependence parameter, a substitution parameter below unity, and a cost elasticity parameter close to (but also slightly below) unity.

IV. Empirical Application: Macro Trends in France

In this section, we combine our theoretical and estimation results to study the predictions of our framework for the response of the French economy to the fall in IT prices. We use the estimation results of Section IIIC to infer the firm-level technological elasticities for all the French firms based on our micro data. We then apply the theoretical results of Section IIB to derive the responses of firm output, prices, and labor share and then to aggregate them to derive the response at the macro level.

As discussed in Section IA, our software and hardware data used in Sections I and III come from the EAE survey and BRN tax files, respectively.⁴² To compute

⁴²The two data sources cover the majority of employment and value added in the French economy and include all firms above certain thresholds of revenue or employment. The EAE (BRN) includes a representative (selective) sample of smaller firms. The combination of RSI and BRN tax files constitutes the universe of all tax-paying firms. See Section OA.2.4 of the online Appendix for more details.

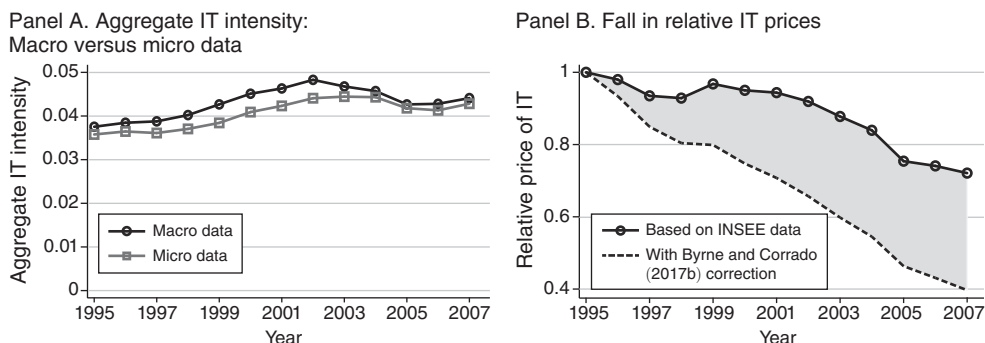


FIGURE 2. THE EVOLUTION OF IT INTENSITY AND PRICES IN FRANCE

Notes: Panel A presents the aggregate IT intensity of the French economy, defined as the ratio of the payments to IT relative to all factor payments, based on INSEE National Account data (macro) and based on the calibrated dataset that maps our micro-level data to the French macroeconomy (micro) (Sources: EU KLEMS/RSI+BRN+EAE Data). Panel B presents the evolution of relative price of IT for France constructed based on the INSEE National Accounts data, and that constructed by incorporating a correction based on the estimates of Byrne and Corrado (2017b) for the bias in the official prices of IT investment goods in the US data.

aggregate technological elasticities, we additionally include in the analysis for this section the remaining, smaller, tax-paying firms, which report to the simplified (RSI) instead of the normal tax regime BRN and are never surveyed by the EAE. For this sample of RSI firms, information on the composition of firm-level investments is not available, so we impute their factor payments based on industry-specific relationships between factor payments, value added, and employment. Our final step to map our micro-level data to the French macro economy is to proportionally adjust the factor payments of all firms within the same industry, in order to match industry-level ratios of factor payments to value added.⁴³ Figure 2, panel A compares the aggregate IT intensity based on the macro INSEE time series, with those constructed based on the aggregate of the resulting extended micro data.

To gauge the magnitude of the fall in IT prices, we also construct a series for the relative price of IT as the ratio of the price of the bundle of IT inputs to the price of the bundle of non-IT inputs, as defined in equation (29). Using the official series on the evolution of investment prices, and depreciation rates by industry and asset type, we construct measures of the rental price (user costs) for software, hardware, and non-IT capital. Combining this information with data on average wages allows us to compute the relative price of IT from 1995 to 2007, the period for which our micro data are available. More specifically, the relative price is given by $W_t \equiv W_{It}/W_{Nt}$, where $W_{It} \equiv (W_{St}/\beta)^\beta [W_{Ht}/(1-\beta)]^{1-\beta}$, $W_{Nt} \equiv (W_{Kt}/\alpha)^\alpha [W_{Lt}/(1-\alpha)]^{1-\alpha}$, and where W_{Lt} is the wage rate and W_{jt} denotes the rental price of factor j . For parameters (α, β) in the two bundles, we use the aggregate shares of non-IT capital and hardware in non-IT factor payments and IT factor payments, respectively (see Section OA.2.2 of the online Appendix for further details).

⁴³ We provide more details about the imputations and this adjustment in Section OA.2.8 of the online Appendix.

Recent work has highlighted the potential inadequacy of the extent of quality adjustment in the official series on the IT investment prices in the United States (e.g., Byrne, Fernald, and Reinsdorf 2016; Byrne and Corrado 2017 a,b; Byrne, Oliner, and Sichel 2018). Since such concerns are likely to be applicable to the French IT prices as well, we apply a correction to our constructed series to account for the potential magnitude of the likely mismeasurement. We first compute the gap between the rate of decline in the estimates of Byrne and Corrado (2017b) for computer prices in the United States and that in the official BEA data. We then add this gap to the rate of decline in our time series based on the INSEE data to construct a *corrected* price series. This approach offers a proxy for the potential magnitude of the unmeasured declines due to quality improvements in the INSEE series.⁴⁴

Figure 2, panel B shows our baseline series for the relative price of IT, which shows a 27.9 percent decline over the 1995–2007 period. It also shows the corrected series that exhibits a much larger decline of 66.7 percent over the same period.⁴⁵

A. Technological Elasticities: From Micro to Macro

In order to apply our aggregation results, we rely on the specification of firm-level technology based on an nhCES production function with the parameter estimates presented in Section IIIC. Using definitions (4) and (11), we find the following relations between the elasticities of cost and marginal cost:⁴⁶

$$(45) \quad \mathcal{E}_{it} = \gamma + \epsilon \Omega_{i,it}, \quad 1 + \mathcal{E}_{it}^m = \gamma + \epsilon \Omega_{i,it}^m,$$

where the pass-through satisfies $\Omega_{i,it}^m = \Omega_{i,it} \left(1 + \left[(1 - \Omega_{i,it}) / \mathcal{E}_{it} \right] (1 - \sigma) \epsilon \right)$, as suggested by equation (9). As we saw, the specification also leads to constant elasticities of factor demand $\sigma_{i,it}$ and $\eta_{i,it}$ from equation (34). All these firm-level technological elasticities depend only on the production function parameters $(\sigma, \gamma, \epsilon)$ and on the firm-level IT intensity $\Omega_{i,it}$ and can thus be computed for every firm i in each year t in our micro data.

Moving on to the aggregate technological elasticities, equations (14) and (16) allow us to express the aggregate elasticities of costs and marginal costs in terms of the firm-level technological elasticities. Due to the linearity of equation (45), these aggregation results together imply the following relationship among the aggregate cost and marginal cost elasticities:

$$(46) \quad \bar{\mathcal{E}}_t = \gamma + \epsilon \bar{\Omega}_{t,t}, \quad 1 + \bar{\mathcal{E}}_t^m = \gamma + \epsilon \bar{\Omega}_{t,t}^m.$$

⁴⁴ Section OA.2.2.4 of the online Appendix provides more details on the procedures used for quality adjustment in France, the comparison between these procedures and those applied by other countries, and additional details on our correction to the price series. Therein, we also discuss an alternative price series that we construct based on the official series in the United States, United Kingdom, and Germany. Section OA.6.4 of the online Appendix reports the results of all the analyses presented in this section using this alternative price series and a corresponding correction.

⁴⁵ As reported in Table OA.13 in the online Appendix, our estimation uncovers negligible trends in IT-biased productivity over this period ($\mu_\theta \approx \mu_\phi \approx 0$). Thus, we take our IT shock to be completely captured by the fall in the relative price of IT.

⁴⁶ See the proof in Section OA.1.2 of the online Appendix.

TABLE 3—AGGREGATE TECHNOLOGICAL ELASTICITIES

Aggregate variable		1995	2007	Average
IT intensity	$\bar{\Omega}_I$	0.036	0.043	0.040
Cost elasticity	$\bar{\mathcal{E}}$	0.954	0.957	0.956
Pass-through	$\bar{\Omega}_I^m$	0.035	0.043	0.040
Marginal cost elasticity	$\bar{\mathcal{E}}^m$	−0.046	−0.043	−0.044
Elasticity of substitution	$\bar{\sigma}_I$	1.066	0.997	1.002
Within contribution		0.235	0.241	0.242
Between contribution		0.831	0.756	0.761
Output elasticity of relative IT demand	$\bar{\eta}_I$	−0.029	0.001	−0.001
Within contribution		0.262	0.268	0.269
Between contribution		−0.291	−0.266	−0.270
Normalized IT intensity variance	ν_I	0.159	0.140	0.137
IT intensity variance	$\text{var}^c(\Omega_{j,i})$	0.005	0.006	0.005
Pass-through variance	$\text{var}^p(\Omega_{j,i}^m)$	0.005	0.005	0.005
IT intensity-cost elasticity covariance	$\text{cov}^c(\Omega_{j,i}, \mathcal{E}_i)$	0.002	0.002	0.002
Pass-through-marginal cost elasticity covariance	$\text{cov}^p(\Omega_{j,i}^m, \mathcal{E}_i^m)$	0.002	0.002	0.002

Note: The table shows the aggregate technological elasticities calculated based on applying our aggregation results to the calibrated dataset, in the first year (1995), the last year (2007), and the average across all years (1995–2007) of the data.

As for the aggregate elasticities of IT demand, Proposition 1 provides us the expressions for the aggregate elasticity of substitution in equation (19) and for the aggregate output elasticity of relative IT demand in equation (20). The two expressions together imply the constraint $\bar{\eta}_{I,t} = (1 - \bar{\sigma}_{I,t})\epsilon$ on the elasticities of aggregate IT demand.⁴⁷ Together, these results allow us to compute all the aggregate technological elasticities based on the firm-level technological elasticities, discussed above, and the distributions of cost Λ_{it}^c , revenue Λ_{it}^r , and pass-through Λ_{it}^p , and the weighted costs Λ_{it}^v . Our data on firm-level factor payments give us the distribution of costs Λ_{it}^c across firms, which we can use to compute the other distributions using equations (15), (17), and (18). Accordingly, we compute the aggregate technological elasticities in each year of the data, assuming the demand elasticity $\lambda = 5$, which corresponds to a value commonly assumed in the literature.

Table 3 reports the results for the first and the final years of our calibrated IT data, as well as the averages across all years. As we saw in Figure 2, panel A, the aggregate IT intensity is close to 4 percent over the period. We also find that the aggregate cost elasticity of the French economy is around 0.96, suggesting a slight degree of increasing returns to scale in the aggregate. Moving to the elasticities of the aggregate marginal costs, the pass-through of IT price to the aggregate price index is

⁴⁷Note that equation (45) implies that $\text{cov}^c(\Omega_{I,it}, \mathcal{E}_{it}) = \epsilon \text{var}^c(\Omega_{I,it})$ and $\text{cov}^p(\Omega_{I,it}^m, \mathcal{E}_{it}^m) = \epsilon \text{var}^p(\Omega_{I,it}^m)$. Substituting these equalities in equation (20) leads to the following result:

$$\bar{\eta}_{I,t} = (1 - \nu_{I,t})(1 - \sigma)\epsilon - \nu_{I,t} \left[\frac{\bar{\mathcal{E}}_t}{1 + \lambda \bar{\mathcal{E}}_t^m} \frac{\text{var}_t^p(\Omega_{I,it}^m)}{\text{var}_t^c(\Omega_{I,it})} - 1 \right] \epsilon = (1 - \bar{\sigma}_{I,t})\epsilon.$$

fairly close to the aggregate IT intensity, and the elasticity of aggregate prices with respect to aggregate output is around -0.04 .⁴⁸

The fifth row of Table 3 shows that the aggregate elasticity of substitution is close to unity, which is in line with the relative stability of the aggregate IT intensity (Figure 2, panel A) despite the falling IT prices (Figure 2, panel B). We can further investigate the within- and between-firm components of the aggregate elasticity of substitution in equation (19). The latter depends on the normalized variance of IT intensity $\nu_{I,t}$ and the ratio of variances of IT intensity and IT pass-through, additionally reported in Table 3. The heterogeneity in IT intensities across firms raises the elasticity of substitution from gross complementarity at the micro level ($\sigma = 0.28$) nearly to unity at the macro level.

In contrast, as we see in the table, the heterogeneity in IT intensities lowers the aggregate output elasticity of relative IT demand $\bar{\eta}_{I,t}$, such that we find scale invariance at the macro level in spite of the presence of strong scale dependence at the micro level. As we saw in Section IIB, the between-firm contribution to the aggregate output elasticity of relative IT demand depends on the degree of covariance between IT intensity and cost elasticity across firms. The table reports this covariance, which is in fact positive due to the fact that firms with higher IT intensity operate with lower degrees of returns to scale, as can be seen in equation (45) in the case of $\epsilon > 0$. As a result, a rise in the scale of aggregate output shifts inputs toward firms with higher degrees of returns to scale, which exhibit relatively lower IT demand. In our data, this mechanism nearly cancels out all the rise in the relative IT demand due to the within-firm contribution, leading to an aggregate output elasticity of relative IT demand close to zero.

B. Accounting for the Contribution of IT to Macro Trends in France

We now use our theoretical results of Section IIB to predict the response of the French economy to the fall in IT prices. These results characterize the general equilibrium response of a given variable at the firm or aggregate levels to a small change in the price of IT at each point in time. For instance, equation (25) gives the general equilibrium response of aggregate output to a small shock in the relative price of IT. Consider now a series for the relative price of IT, $W_{I,t}$, e.g., one based on the official INSEE statistics shown in Figure 2, panel B. Starting with the initial period $t_0 = 1995$, we can predict the cumulative response of any aggregate variable to the IT price shock at each period $t > 1995$ by chaining the response to year-on-year change in all the preceding periods. For instance, to predict the response of the aggregate output $\bar{Y}_t$, we use

$$(47) \quad \log\left(\frac{\bar{Y}_t}{\bar{Y}_{t_0}}\right) \approx \sum_{\tau=t_0}^{t-1} \frac{d\log \bar{Y}_\tau}{d\log W_{I,\tau}} \times \Delta \log W_{I,\tau},$$

where we know the elasticity of aggregate output to IT prices $d\log \bar{Y}_\tau / d\log W_{I,\tau}$ from equation (25). Table 4 shows the results of applying this approach to different

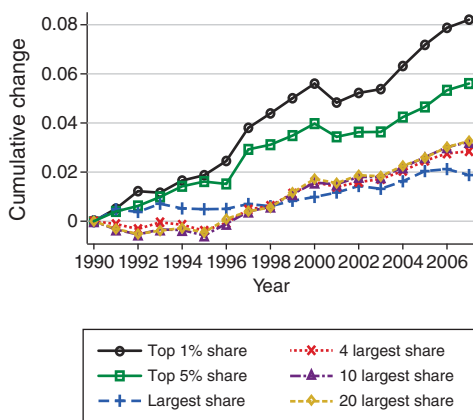
⁴⁸Note that the cost superelasticity $\bar{\epsilon}^s = \bar{\epsilon}^m - (\bar{\epsilon} - 1)$ is very close to zero, implying that the degree of returns to scale does not vary in the scale of aggregate output.

TABLE 4—AGGREGATE-LEVEL CHANGES IN FRANCE, 1995–2007: DATA VERSUS MODEL PREDICTIONS

Change in aggregate variable		Data	Model (INSEE)	Model (correction)
Relative IT price (%)	W		−27.9	−60.3
Aggregate output (%)	$\bar{Y}$	19.1	1.4	3.9
Price index (%)	$\bar{P}$	−15.7	−1.4	−3.7
Share of top 1 percent of firms (pp)		6.3	1.5	4.1
Share of top 5 percent of firms (pp)		4.0	1.3	3.3
Labor share (pp)	$\bar{F}S_L$	0.3	−0.0	−0.0
Within contribution (pp)		1.3	0.7	1.8
Between contribution (pp)		−1.0	−0.7	−1.8
Profit share (pp)		0.0	0.0	0.0

Notes: The table shows the changes in the French aggregate output, its price (relative to the bundle of non-IT inputs), and the profit share (based on the official INSEE series), and in the concentration of production and labor share (based on our macro data) over the 1995–2007 period. The changes are compared with those predicted by the model (based on the calibrated dataset) in response to two different series for falling relative IT prices: baseline series based on INSEE data and that including the correction by Byrne and Corrado (2017b). “pp” stands for percentage points. Percentage changes are expressed relative to the respective baseline in each model and in the data.

Panel A. Production concentration in the data



Panel B. Response of concentration to IT shock

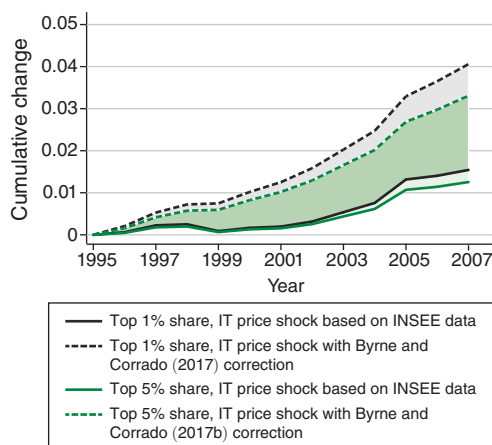


FIGURE 3. RISE OF CONCENTRATION IN FRANCE: DATA VERSUS MODEL PREDICTIONS

Notes: Panel A presents the evolution of the sales-weighted averages (across three-digit industries) of the cumulative change in concentration, measured as the share in total industry sales of the largest 1 percent, 5 percent, 1, 4, 10, or 20 firms. Panel B shows the cumulative changes in the shares of the largest 1 percent and 5 percent of firms predicted by the model (based on the calibrated dataset) in response to the two series for the relative IT prices over the 1995–2007 period.

aggregate variables, including output. The model predicts that the fall in IT prices calculated based on the official INSEE statistics (with the Byrne and Corrado 2017b correction) in the 1995–2007 period is predicted to lead to a 1.4 percent (3.9 percent) rise in the aggregate output of the French economy.⁴⁹

⁴⁹This result suggests that the fall in IT prices can explain between 7 and 23 percent of the overall 19 percent rise in output per worker in France between 1995 and 2007.

We next examine the trends in the concentration of production in France. Figure 3, panel A shows a sizable increase in different proxies of industrial concentration of production, averaged across three-digit industries, starting in 1990.⁵⁰ To investigate the predictions of our model about this trend, note that equation (28) characterizes the general equilibrium response of the market share of each firm in the data to falling IT prices, given the response in the aggregate output computed above. Using this result, we find the predicted change in the concentration of production in the French economy using our micro data starting in 1995, the first year in which our IT data become available. Figure 3, panel B shows the resulting predicted cumulative change in the shares of the top 1 percent and 5 percent of firms in the data based on our model.⁵¹ Since larger firms are more IT intensive, the model predicts a rise in the concentration of production. Table 4 shows that, depending on the measure of concentration and the magnitude of the IT shock considered, the model explains roughly between one- to three-quarters of the rise observed in the data over the 1995–2007 period.

Next, we examine the evolution of the labor share in France over the period 1990–2007. Figure 4, panel A shows the cumulative change in the aggregate wage bill (including payroll taxes) as a share of aggregate value added for our sample of all tax-paying French firms. The stability of the aggregate labor share masks sizable compositional changes. Following the strategy used recently by Kehrig and Vincent (2021) for the US data, we further decompose the changes in the aggregate labor share into the contributions of within-firm change in labor share and the cross-firm reallocations. To find the two contributions, we compute the aggregate change first keeping fixed the share of firms in aggregate value added and then keeping fixed the firm-level labor shares.⁵² Figure 4, panel A shows a divergence in France between the within-firm and between-firms contributions to the aggregate labor share: the former rose within this period, while the latter fell. This is in line with the findings of Autor et al. (2020) and Kehrig and Vincent (2021) in the case of the United States.⁵³

We now study the predictions of the model about the response of the aggregate labor share, and the decomposition above, to the fall in relative IT prices. First, note that given the assumptions in equation (29), the labor share in our model is a constant fraction of the share of non-IT factors, that is, $FS_{L,it} = (1 - \alpha)FS_{N,it}$ and $\bar{FS}_{L,t} = (1 - \alpha)\bar{FS}_{N,t}$. We can thus use equation (26) to find the response of the aggregate labor share to the change in relative prices in each period, which we can chain over time using a similar approach to that used in equation (47). This allows us to compute the cumulative response of the aggregate labor share to falling IT prices in our data starting in 1995. Figure 4, panel B shows the resulting series. In line with the data, the model predicts the aggregate labor share to be unresponsive to falling IT prices.

⁵⁰For these results (and those reported in Figure 4, panel A), we can extend our data farther back in time to 1990 since they do not require information on firm-level IT investments, which is available only starting in 1995.

⁵¹We focus on these two measures of concentration since they do not rely on the granularity of firms and are thus compatible with our assumption of continuum of firms in Section IIB.

⁵²Section OA.6.1 in the online Appendix provides the full details of this decomposition exercise.

⁵³Figure 4, panel A shows that the within-firm component of the aggregate labor share shows a sizable drop of around 1 percentage point in the last year of our data, from 2006 to 2007. To avoid this confound, Table 4 reports the cumulative change in the aggregate labor share and its components from 1995 to 2006.

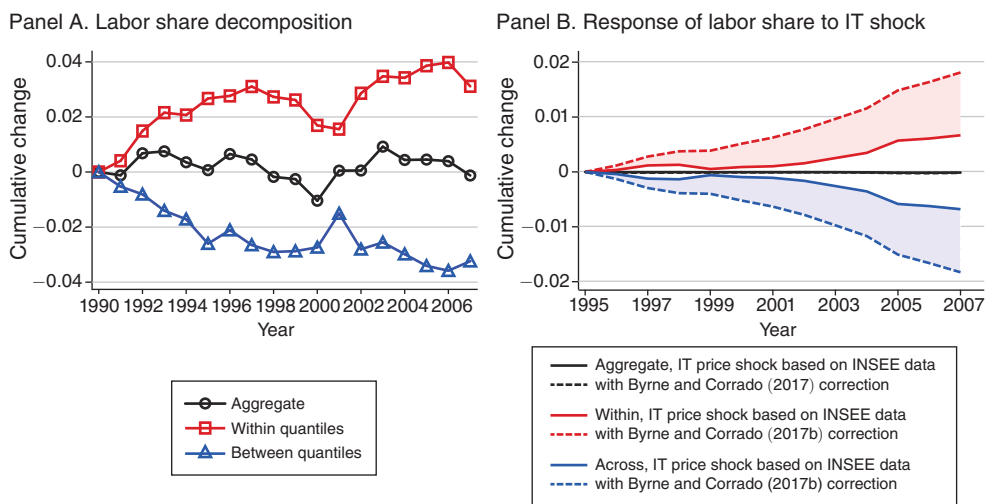


FIGURE 4. EVOLUTION OF LABOR SHARE IN FRANCE: DATA VERSUS MODEL PREDICTIONS

Notes: Panel A presents the cumulative change in the aggregate labor share, as well as the decomposition of this change to within and across-firm components (at the level of two-digit industries). Panel B shows the corresponding responses predicted by the model (based on the calibrated dataset) to the two series for the relative IT prices over the 1995–2007 period.

We can further use equation (28) to compute the contribution of shifts in market shares across firms to the aggregate labor share at each point in time.⁵⁴ Figure 4, panel B illustrates the resulting predictions about the cumulative contribution of between-firm reallocations over this period. As the fall in IT prices leads to a reallocation toward large and IT-intensive firms, we find a negative contribution to the aggregate labor share due to the fact that these firms typically pay a smaller share of their income to labor. From equation (5), we know that the share of non-IT inputs in firm income is given by $FS_{N,it} = \Omega_{N,it}/\mu\mathcal{E}_{it}$. In the cross section of firms, large and IT-intensive firms have lower labor shares both because they have higher IT intensities (lower $\Omega_{N,it}$) and because they have higher cost elasticities $\mathcal{E}_{it}$.⁵⁵ Finally, the rise in the labor share simply follows from the fact that, in response to falling IT prices, the share of non-IT inputs in firm income rises due to gross complementarity (substitution channel), and, as stated in Lemma 1, the cost elasticity of the firm falls due to the positive firm-level output elasticity of relative IT demand (returns to scale channel).

Finally, Table 4 reports that, once we account for the shares of both labor and capital in aggregate income, the data suggest a negligible change in the aggregate profit

⁵⁴To see this, note that we have $\bar{F}S_{N,t} = \sum_i \Lambda_{it}^r \cdot FS_{N,it}$, and thus, the between-firm contribution to the changes in the aggregate labor share is given by $\sum_i \Lambda_{it}^r \cdot FS_{N,it} \cdot d\log \Lambda_{it}^r / d\log W_{1,t} \cdot d\log W_{1,t}$.

⁵⁵Figure OA.2, panel B in Section OA.5.1 of the online Appendix shows that this negative relationship between firm size and labor share holds in our data and that our estimated production function in fact explains a sizable part of the observed relationship in the data.

share.⁵⁶ Using equation (27), we can compute the predicted year-on-year responses and, again, accumulate them over time in line with the approach in equation (47). Despite the fact that the aggregate profit share is endogenous, our results suggest a negligible response to the IT price shock. This is evident in the results reported in Table 3, where we find a small aggregate output elasticity of IT demand $\bar{\eta}_{I,t}$ and aggregate cost superelasticity $\bar{\mathcal{E}}_t^m - (\mathcal{E}_t - 1)$ in our data.

Overall, the results of this section establish that the predictions of our framework about the response of the French economy, in terms of concentration and labor and profit shares, to falling IT prices are qualitatively and quantitatively in line with the observed data.

V. Discussion

A. Why Would IT Demand Be Scale Dependent?

In this paper, we empirically documented the connection between firm scale and relative IT demand and studied the implications of this aspect of firm-level technology for the aggregate economy. For the most part, we have not taken a stance on the underlying drivers of such a relationship *within* the firm. Multiple alternative explanations for the emergence of this relationship exist. A larger scale of production may require a deeper internal hierarchical structure for the firm, requiring relatively more IT per worker to allow additional coordination across workers at different levels of the hierarchy (Garicano and Rossi-Hansberg 2006; Caliendo and Rossi-Hansberg 2012; Mariscal 2018). Moreover, it may also require increasing the number of establishments, the number of markets served, or the number of products produced. In each case, we can make the case that the coordination needs of the firm rise more than proportionally with the size of employment and output, potentially leading to scale-dependent IT demand. In line with this insight, Section OA.3.1.5 of the online Appendix presents evidence for a positive relation between IT intensity and the number of establishments, the depth of the hierarchical structure, and the number of exporting products and destinations.

Our nhCES parameterization of firm-level technology in Section IIIA provided a stylized and reduced-form account of these different margins. We provided a motivation for this specification based on the idea that IT inputs particularly enhance the productivity of a firm-specific organizational factor that otherwise exhibits decreasing returns. In fact, a sizable body of empirical work has documented such connections between IT and the organizational efficiency of the firm. Brynjolfsson and Hitt (2000) discuss numerous case studies that demonstrate the effect of IT on the organizational practices of businesses. A variety of firm-level cross-sectional studies have also documented the complementarity between IT adoption and organizational capital (e.g., Bresnahan, Brynjolfsson, and Hitt 2002; Brynjolfsson and Hitt 2003; Baker and Hubbard 2004). In particular, Bartel, Ichniowski, and Shaw (2007) use

⁵⁶By definition, the share of economic profits is one minus the sum of the income shares of labor and capital (the latter inclusive of IT). Section OA.6.3 of the online Appendix discusses how we construct our measures of capital share in the French economy that allow us to compute the aggregate profit share. As with the labor share, we also find the aggregate share of capital to be stable in the 1990–2007 period.

detailed firm-level data on setup times, run times, and inspection times to show how IT improves the efficiency of organizing multiple production processes within the firm. As another example, Bloom et al. (2014) find that the adoption of IT impacts the organization of the firm in terms of span of control over individual workers. In sum, IT allows firms to more effectively apply their technology or core competencies to a larger scale of inputs, across a wider range of activities, and to serve more buyers and markets.

We note that the assumption of decreasing returns to a firm-specific organizational input is by no means novel.⁵⁷ At least since Williamson (1967), many economic theories have provided microfoundations for such firm-level organizational limits to scale.⁵⁸ These models offer various mechanisms through which the complexities of coordination and communication, both within the firm and also between the firm and its buyers and suppliers, may rise as the firm scale grows. A larger scale of output brings about complexities that stem from organizing production tasks over larger scales of inputs and outputs. On the input side, the firm potentially has to find and coordinate with more intermediate suppliers, hire more or different types of workers, procure more machines, and manage across a larger set of inputs. On the output side, it has to potentially manage larger inventories, coordinate with more buyers or across more markets, provide support services to a larger set of customers, organize larger-scale marketing efforts, and solve larger delivery or distribution problems. Again, the idea is that IT is an input that specifically enhances the productivity of firms in fulfilling such management and coordination.

B. Technology or Market Power?

In this paper, we studied the implications of a fall in the price of IT inputs under any smooth specification of firm-level technology. As such, our approach accounts for any technological mechanism through which the utilization of IT affects production at the firm level. This approach contrasts from some of the recent work that considers a specific technological mechanism, e.g., the effect of IT on the organizational hierarchy (Mariscal 2018), on costs of spanning multiple markets (Aghion et al. 2023), or on the trade-off between fixed and marginal costs (De Ridder 2024). By construction, all such technological forces at the firm level should manifest themselves in terms of one or several of the technological elasticities considered in Section IIA.

⁵⁷ Considering organization as a firm-specific input also has a long history in economic thought. In his *Principles of Economics*, Marshall (1920) identifies organization as a fourth factor of production, in addition to labor, capital, and land. Coase (1937) argues that decreasing returns to the entrepreneur function is a key factor in determining the boundary of the firm. To justify firm-level decreasing returns, McKenzie (1959) defines a firm-specific entrepreneurial input, Lucas (1978) refers to it as the managerial talent of the firm's manager, and Atkeson and Kehoe (2005) interpret it as the "organizational capital" of the firm. Our concept of organizational input is also in line with the concept of the firm-level economic competencies that is considered to be a core component of the intangible capital of firms (Haskel and Westlake 2018).

⁵⁸ Some approaches emphasize the agency issues inherent in delegation and the formation of hierarchies (see Mukherjee 2012 for a review), while others focus on the complexities of solving allocative decisions within the firm (see Garicano and Van Zandt 2013). In particular, in theories of knowledge hierarchies, the complexity of the process of dealing with production errors changes endogenously with the scale of operation (Garicano 2000; Garicano and Rossi-Hansberg 2006; Caliendo and Rossi-Hansberg 2012). Another alternative follows the approach of Simon (1962) and examines the implications of bounded rationality and the limits to information processing (e.g., Van Zandt and Radner 2001).

Our theory shows that, due to the presence of the returns to scale channel studied here, IT can in principle lead to a rise in concentration and/or the profit share, while still bringing about substantial productivity gains. In contrast, a sizable literature has emerged that interprets the trends in concentration, profits, and labor share through the lens of variations in markups and market power (e.g., De Loecker, Eeckhout, and Unger 2020; Baqaee and Farhi 2020; Aghion et al. 2023). Inspired by this literature, work on the implications of IT on these trends has often considered a combination of a specific technological mechanism, as discussed above, along with its additional impacts on market power. To conceptually distinguish between the efficient technological mechanisms and the distortionary mechanisms involving market power, we focus on delineating the technological aspects of the aggregate effects of IT in a setting involving efficient allocations across firms.

Nevertheless, our framework can also be used to jointly study a comprehensive account of the technological effects of IT along with the effects on market power. Consider, for instance, using an alternative demand aggregator such as Kimball preferences instead of the CES demand for firm products. This alternative specification allows for heterogeneity in markups across firms and can be calibrated to match the positive relationship typically considered between markups and firm size (see, e.g., Edmond, Midrigan, and Xu 2023). We can rely on our results to draw insights about the response of this modified model to the rise of IT within and across firms. In particular, the model predicts that falling IT prices lead to the rise of output of large firms and a decline among small firms. This leads to a rise in markups for large firms and a fall for small firms.⁵⁹ In addition, the resulting cross-firm reallocations lead to a shift of market shares toward high-markup firms and away from low-markup firms. These patterns are in line with those documented by De Loecker, Eeckhout, and Unger (2020). We believe exploring the predictions of this richer model for the relative size of the effects of IT on technology and market power could be a fruitful avenue for future work.

Lastly, we note that our results may have implications for the current approaches to the estimation of markups and their variations across firms following De Loecker and Warzynski (2012). This approach relies on the estimation of output elasticities of a given variable input, e.g., labor in our setting, in order to infer markups. As emphasized recently by Demirer (2020), accounting for factor-augmenting productivity—IT-biased productivity in our setting—is important to account for the endogenous variations in output elasticities. In our quantitative model, the output elasticity of labor is given by $(1 - \alpha)[(1 - \Omega_{L,i})/(\gamma + \epsilon \Omega_{L,i})]$, which is decreasing in IT cost share $\Omega_{L,i}$ and therefore firm size. To the extent that the presence of scale dependence ($\epsilon > 0$) intensifies the negative relationship between size and the output elasticity of labor, it may weaken the relationship between size and the implied markup.

⁵⁹We note, however, that the rise (fall) of markup among large (small) firms in turn curbs the first-order effect of the rise of IT on output. In other words, under preferences with variable price elasticities, the output of large (small) firms rises (falls) less than that under our benchmark with constant price elasticities.

VI. Conclusion

In this paper, we presented a theory that characterizes the technological aspects of the response of the macroeconomy to a shock to the price of a given factor of production. Allowing for any smooth technology at the firm level, our theory shows that potential nonhomotheticity in production (i.e., the dependence of factor demand on firm scale) leads to the endogenous response of returns to scale to falling factor prices and generates a host of new channels affecting the macroeconomy. Applying the theory to understand the aggregate consequences of falling IT prices, we empirically examined the extent of scale dependence in IT demand at the firm level. Relying on novel data on the investment and capital stocks of firms in software and hardware in the universe of French firms, we found that the intensity of IT demand strongly and robustly rises with firm size. We then estimated a nonhomothetic firm-level production function using our micro data and used the resulting estimates to calibrate a quantitative model based on our theory. The model predicts, just as we find in the data, that the observed fall in the price of IT in France results in a reallocation of market shares across firms toward those firms with larger IT intensity and size and lower levels of returns to scale and labor share. These reallocations cancel out the positive within effects of IT on returns to scale and labor share but also generate additional productivity effects in the aggregate.

APPENDIX A. THEORETICAL APPENDIX

A. Cost Superelasticity

While the cost superelasticity cannot be fully expressed in terms of cost and factor demand elasticities (Ω_j , $\mathcal{E}$, σ_j , and η_j), it still (generically) depends on the output elasticity of relative factor demand η_j . To see this, let $\mathbf{W}_{-j}$ stand for the vector of all factor prices other than factor j and assume that the elasticity of substitution is not unity $\sigma_j \neq 1$. Since the factor intensity $\Omega_j(Y; \mathbf{W})$ is a monotonic function of factor price W_j , we can define a transformation $\hat{\mathcal{E}}$ of the cost elasticity function such that $\hat{\mathcal{E}}(Y; \Omega_j(Y; \mathbf{W}), \mathbf{W}_{-j}) \equiv \mathcal{E}(Y; \mathbf{W})$. Equations (6) and (8) together imply

$$(A1) \quad \frac{\partial \hat{\mathcal{E}}(Y; \Omega_j, \mathbf{W}_{-j})}{\partial \Omega_j} = \frac{\partial \mathcal{E} / \partial \log W_j}{\partial \Omega_j / \partial \log W_j} = \frac{\eta_j}{1 - \sigma_j}.$$

Equation (A1) has an important implication for the cross-sectional relationship between the income shares and the IT cost share. Let us consider the empirically relevant case of $\eta_i > 0$. We know that the revenue-to-cost ratio is given by $\mu \mathcal{E}_i$. If IT and non-IT inputs are gross complements ($\sigma_i < 1$), then the cost elasticity is increasing in the IT cost share, implying a positive relationship between the IT cost share and the profit share.

If we additionally impose the restriction that the cost elasticity only depends on the IT cost share, $\mathcal{E} = \hat{\mathcal{E}}(\Omega_j)$, and define the cost superelasticity as the derivative of cost elasticity with respect to output, we find

$$(A2) \quad \mathcal{E}^s \equiv \frac{\partial \log \mathcal{E}(Y; \mathbf{W})}{\partial \log Y} = \Omega_j \frac{1 - \Omega_j}{\mathcal{E}} \frac{\eta_j^2}{1 - \sigma_j}.$$

The cost superelasticity is important for two reasons. First, it determines the marginal cost elasticity $\mathcal{E}^m$ in equation (11) through $\mathcal{E}^m = \mathcal{E} - 1 + \mathcal{E}^s$. Second, it informs us about the relationship between returns to scale and the scale of output: if $\eta_j > 0$ and if factor j is gross complements with other factors in the sense that $\sigma_j < 1$, then the cost elasticity is increasing in output, implying there is a positive relationship between output and the profit share. Note that in the example of the nhCES production function of Section IIIA, the cost superelasticity satisfies $\mathcal{E}_{it}^s = \epsilon(\Omega_{I,it}^m - \Omega_{I,it})$.

B. Comparative Statics of Firm-Level Allocations

The following lemma characterizes the reallocations of output across firms as a result of the change in a factor price or in the scale of aggregate output.

LEMMA 3: *The first-order responses in the allocation of output across firms to shocks in the price of a factor input or aggregate output are given by*

$$(A3) \quad \frac{\partial \log \mathcal{Y}_i(\bar{Y}; \mathbf{W})}{\partial \log W_j} = -\lambda \frac{\Omega_{j,i}^m - \bar{\Omega}_j^m}{1 + \lambda \mathcal{E}_i^m}, \quad \frac{\partial \log \mathcal{Y}_i(\bar{Y}; \mathbf{W})}{\partial \log Y} = \frac{1 + \lambda \bar{\mathcal{E}}^m}{1 + \lambda \mathcal{E}_i^m},$$

where the aggregate pass-through $\bar{\Omega}_j^m$ and the aggregate marginal cost elasticity $\bar{\mathcal{E}}^m$ satisfy equation (16).

PROOF:

See Section OA.1 of the online Appendix.

Lemma 3 shows that the reallocation response is characterized by the firm-level elasticities of the marginal cost $\Omega_{j,i}^m$ and $\mathcal{E}_i^m$, defined in equation (11). Recall that assuming a production function with a degree of returns to scale $\mathcal{E}_i$ that is constant and homogeneous across firms, we have $\Omega_{j,i}^m = \Omega_{j,i}$ and $\mathcal{E}_i^m = \mathcal{E} - 1$. In this case, the response of firm-level output with respect to the price of factor j is given by the difference between the firm-level and the aggregate share of the factor in costs, $-\lambda(\Omega_{j,i} - \bar{\Omega}_j)$, and the elasticity of firm-level output to the aggregate output shock is equal to one for all firms. Once we allow for the heterogeneity in returns to scale across firms, two main differences emerge in equation (A3). First, the pass-through to marginal cost $\Omega_{j,i}^m$ deviates from the factor intensity $\Omega_{j,i}$ following equation (9). Second, the response of output is weaker for firms with higher marginal cost elasticity $\mathcal{E}_i^m$. The distribution of pass-throughs Λ_i^p provides a weighting of firms that underweights firms with higher marginal cost elasticities, whose outputs are less reactive to changes in factor prices or aggregate output.

C. Microfoundation for Firm-Level Technology

Assume that the firm i has access to a collection $\mathcal{I}_i$ of different alternative recipes to produce output Y_i by combining non-IT input X_{Ni} and IT input X_{Ii} . Each recipe ω describes a procedure to make output y_ω by combining $a_\omega y_\omega^{\mathcal{E}_\omega}$ units of non-IT inputs and $b_\omega y_\omega^{\mathcal{E}_\omega}$ units of IT inputs, where $(a_\omega, b_\omega, \mathcal{E}_\omega)$ is a tuple of recipe-specific exogenous constants that satisfy $a_\omega, b_\omega > 0$ and $\mathcal{E}_\omega > 1$.⁶⁰ We thus have

$$Y_i = \int_{\mathcal{I}_i} y_\omega d\omega, \quad X_{Ni} = \int_{\mathcal{I}_i} a_\omega y_\omega^{\mathcal{E}_\omega} d\omega, \quad X_{Ii} = \int_{\mathcal{I}_i} b_\omega y_\omega^{\mathcal{E}_\omega} d\omega.$$

Note that the recipe-level production functions do not allow for substitution between the two factors ($\sigma_\omega = 0$), are homothetic ($\eta_\omega = 0$), and have cost elasticity $\mathcal{E}_\omega$. Thus, we have $\Omega_{I,\omega}^m = \Omega_{I,\omega} \equiv (b_\omega W)/(a_\omega + b_\omega W)$ and $\mathcal{E}_\omega^m = \mathcal{E}_\omega - 1$. Moreover, the outputs of different recipes are perfectly substitutable.

To characterize the elasticities of the firm-level cost function, we can apply Proposition 1 to find the elasticities of the cost function for the firm, in the limit of $\lambda \rightarrow \infty$ due to perfect substitutability of output of different recipes. The following corollary of the proposition characterizes the firm-level elasticities of factor demand.

COROLLARY 1: *Let $\Lambda_{i,\omega}^r$ denote the distribution of output across different production recipes ω of firm i , and let $\Lambda_{i,\omega}^c \propto (1/\mathcal{E}_\omega)\Lambda_{i,\omega}^r$ and $\Lambda_{i,\omega}^p \propto [1/(\mathcal{E}_\omega - 1)]\Lambda_{i,\omega}^r$ denote the distributions of costs and pass-throughs across these recipes. Then, the firm-level elasticity of substitution and the output elasticity of relative IT demand satisfy*

$$(A4) \quad \sigma_{I,i} = \frac{\mathcal{E}_i/\mathcal{E}_i^m}{\Omega_{I,i}(1 - \Omega_{I,i})} \text{var}_i^p(\Omega_{I,\omega}), \quad \eta_{I,i} = \frac{\mathcal{E}_i^m}{\Omega_{I,i}(1 - \Omega_{I,i})} \text{cov}_i^c\left(\Omega_{I,\omega}, \frac{1}{\mathcal{E}_\omega - 1}\right),$$

where, as before, we have defined the firm-level IT cost share $\Omega_{I,i} \equiv E_i^c[\Omega_{I,\omega}]$, cost elasticity $\mathcal{E}_i \equiv E_i^c[\mathcal{E}_\omega]$, and marginal cost elasticity $\mathcal{E}_i^m \equiv E_i^p[\mathcal{E}_\omega] - 1$, and where E_i^x , var_i^x , and cov_i^x denote the expectation, variance, and covariance operators, respectively, with respect to distribution $\Lambda_{i,\omega}^x$.

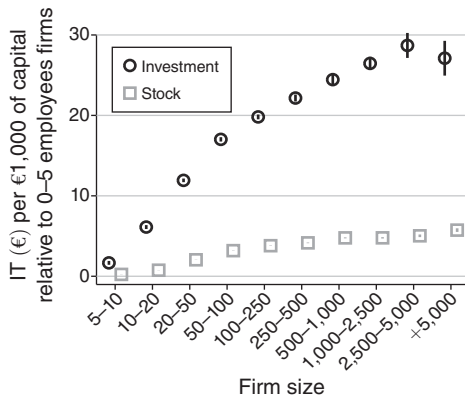
PROOF:

See Section OA.1 of the online Appendix.

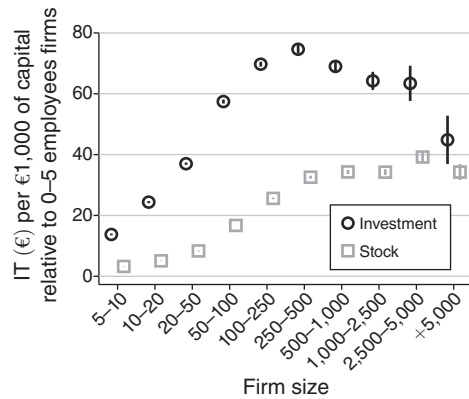
Equation (A4) shows two distinct microfoundations for the elasticity of substitution and the output elasticity of relative IT demand. The former emerges due to heterogeneity in the IT intensity across different production recipes, whereas the latter emerges due to a correlation between IT intensity and returns to scale across different production recipes.

⁶⁰ For an alternative microfoundation in which recipes have constant marginal costs but are heterogeneous in terms of a fixed cost, see Section OA.4.1.3 of the online Appendix.

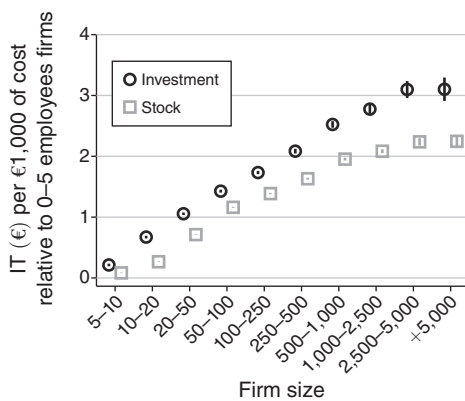
Panel A. Software per unit of capital



Panel B. Hardware per unit of capital



Panel C. Software per unit of costs



Panel D. Hardware per unit of costs

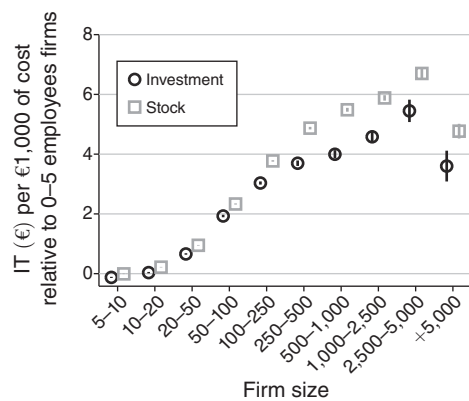


FIGURE B1. CROSS-SECTIONAL RELATIONSHIP BETWEEN IT AND FIRM SIZE

Notes: This figure reports average IT investment and stock per €1,000 of capital and cost by firm size class. In panels C and D, Investment and Stock stand for measures based on investment values and factor payment, respectively. Averages are conditional on a set of flexible fixed effects constructed from the interaction of three-digit industry codes and time dummies, and a full set of cohort fixed effects (pre-1980, 1980–1993, 1993–1995, ..., 2005–2007) and normalized age fixed effects. In the case of software, sample is all firms that were sampled in EAE (that year for investment, at least once for capital). In the case of hardware, sample is all firms that reported hardware investment lower than 0.99 times total investment. The bands around the estimates show the 90 percent confidence intervals.

APPENDIX B. ADDITIONAL DETAILS ON EMPIRICAL RESULTS

A. Other Proxies of Relative IT Demand

Figure B1 complements Figure 1 in the main text and examines the cross-sectional relationship between firm size and IT demand relative to capital or to costs.⁶¹ First, panels A and B estimate equation (1) replacing IT per employee on the left-hand side by the ratio of the firm's IT investment (or stock) to the total capital investment

⁶¹ For further details on the construction of all our measures of relative IT demand, see Section OA.3.1.1 of the online Appendix.

TABLE B1—REGRESSIONS OF IT PER EMPLOYEE ON LOG FIRM SIZE, BY BINS OF EMPLOYMENT

	IT per employee				
	[1;50[	[50;100[	[100;250[	[250;1,000[	≥1,000
<i>Panel A. Software (stock)</i>					
Size (proxied by sales)	0.3964 (0.0057)	0.5160 (0.0137)	0.4490 (0.0141)	0.4755 (0.0190)	0.3768 (0.0396)
Observations	393,470	94,433	67,577	31,031	6,154
R^2	0.259	0.232	0.250	0.306	0.391
Size (proxied by VA)	0.3306 (0.0064)	0.5589 (0.0168)	0.4845 (0.0172)	0.5233 (0.0219)	0.4041 (0.0425)
Observations	393,527	94,455	67,585	31,038	6,154
R^2	0.255	0.229	0.248	0.305	0.391
<i>Panel B. Hardware (stock)</i>					
Size (proxied by sales)	0.1685 (0.0009)	1.0493 (0.0063)	0.8140 (0.0061)	0.5302 (0.0074)	0.3255 (0.0137)
Observations	2,645,249	137,207	95,862	41,973	8,236
R^2	0.446	0.453	0.502	0.510	0.579
Size (proxied by VA)	0.0688 (0.0010)	0.9227 (0.0078)	0.7437 (0.0074)	0.5045 (0.0087)	0.2933 (0.0143)
Observations	2,645,384	137,244	95,889	41,994	8,236
R^2	0.440	0.402	0.466	0.489	0.570

Notes: In both panels the dependent variable is the logarithm of IT stock per employee. Standard errors are reported in parentheses. In columns 1–5 we report results of regressions for firms in various bins of total number of employees: fewer than 50 employees, 50 to 100, ... up to more than 1,000 employees. The independent variable is the logarithm of firm size either proxied by sales or value added. The time period is 1995–2007. In panel A the sample is all firms sampled by EAE, and in panel B the sample is BRN firms. All columns include a full set of three-digit industry classification fixed effects interacted with year fixed effects and a full set of cohort fixed effects (pre-1980, 1980–1993, 1993–1995, ..., 2005–2007) and normalized age fixed effects. An elasticity of 0.3997 means that raising sales by a factor of 2 raises IT per employee by 39.97 percent.

(or stock).⁶² Second, panels C and D show the ratio of IT investments to the sum of the firm's wage bill and total investments (Investment measure) and the ratio of the payments to software or hardware divided by the sum of payments to all production factors (Stock measure). Payments to capital factors are computed as the product of the user cost for each type of capital and the nominal value of the corresponding stock.⁶³

B. Monotonicity of the Size-IT Relation

A number of potential issues may complicate the interpretation of our reduced-form evidence in Section IB. First, we may be concerned that small firms face some fixed cost of adopting IT, may not perfectly report their IT investment, or may face different costs of IT compared to large firms in our data. Second, we may be concerned that

⁶² Section OA.3.1.4 of the online Appendix discusses the reasons behind the seemingly nonmonotonic pattern observed in the case of hardware per unit of capital in Figure B1.

⁶³ The user cost is the sum of the long-term interest rate on government bonds, the depreciation rate specific to each type of capital, and the expected fall in the price of that type of capital, computed as the three-year moving average of the investment price. See Section OA.2.2 of the online Appendix for more details.

small firms covered in our data are not representative of all small firms in the French economy. To address these concerns, we show that the relationship between firm size and relative IT demand also appears among large firms in our data. Figures 1 and B1 already show that the positive relationship between size and IT intensity is fairly consistent across different brackets of size, particularly in the case of software, where the issues of selection and measurement error are less pronounced.⁶⁴ Moreover, our reduced-form IV identification results, presented in Section OA.3.3 of the online Appendix, confirm that the relationship between firm size and IT intensity also appears among exporting firms, which tend to be large manufacturing firms.

To address this issue more directly, Table B1 replicates the within-industry results of Table 1 (in terms of IT per employee) for samples of firms in different brackets of employment size. The results for software per employee clearly demonstrate that the relationship between size and relative IT demand is fairly robust across different brackets of firm size. Again, the results show more variability in the case of hardware data, but the coefficients are still nonmonotonic, first increasing and then decreasing.

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⁶⁴ As we already mentioned, the hardware data include office furniture investments, which biases our measure of hardware intensity upward, particularly for middle-size firms for which office capital may constitute the largest share of their capital stock. See Section OA.3 of the online Appendix for a discussion of how this issue appears in Figure 1. With regard to selection, note that the source of our hardware data includes firms that have voluntarily chosen to file in the BRN tax regime below certain revenue thresholds. It is likely that small firms selecting to file in the BRN regime expect higher future growth, which may also be correlated with currently higher IT intensities. This would make small firms in the BRN sample unsuitable representatives for the sample of all small French firms. In contrast, the source of our software data, the EAE dataset, contains a representative sample for firms with fewer than 20 employees.

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